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European Network Of Public Employment Services

PES support to green skills and jobs: From commitment to practical steps and strategies

Toolkit



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INTRODUCTION



What is this Toolkit about?

A greener economy will require the development of new skillsets as well as support for workers, jobseekers and employers to manage job-to job transitions. Thus, most importantly, the green transition is expected to have a major impact on changing skills requirements[1].

This Toolkit supports Public Employment Services (PES) to provide practical guidance and promote skills development for the green transition. PES can play a significant role in identifying and anticipating skills needs, increasing matching efficiency for green jobs, to support job-to-job transitions into green and green-driven occupations, to equip workers with demanded skills, and to ensure the green transition is fair. The Toolkit aims to strengthen PES' operational capacities to effectively support when implementing fair transition policies, with a focus on labour market transitions and improving skills intelligence.

The political guidelines by the European Commission 2024-2029[2] reinforce the goals of the European Green Deal, namely that Member States have pledged reduce net greenhouse gas (GHG) emissions by at least 55% by 2030, compared to 1990 levels and the European Commission has adopted a set of proposals to make the EU's climate, energy, transport and taxation policies. One-third of the EUR 1.8 trillion investments from the NextGenerationEU Recovery Plan and the EU's seven-year budget will finance the European Green Deal[3]. Contributing to the Green Deal features among the objectives of the Decision establishing the PES Network[4].

The Council Recommendation on ensuring a fair transition towards climate neutrality[5] stresses that the transition needs to be fair and calls on Member States to 'strengthen the operational capacities of the relevant public services to provide effective guidance and support for implementing fair transition policies; in particular, strengthen public employment services to support labour market transitions and skills intelligence'.

The European Pillar of Social Rights Action Plan[6] highlights that unity, coordination and solidarity are needed to move towards a greener decade. The green transition is also facilitated by several EU financial instruments, such as the Just Transition Mechanism[7] (see Section 3), the Social Climate Fund[8], and the European Social Fund

1 European Network of Public Employment Services, Greening of the labour market – impacts for the Public Employment Services, Small-scale study, 2021, available https://op.europa.eu/en/publication-detail/-/publication/a5ce471b-f0dd-11eb-a71c-01aa75ed71a1/language-en.

2 Political guidelines for the next European Commission 2024–2029, https://commission.europa.eu/document/download/e6cd4328-673c-4e7a-8683-f63ffb2cf648_en?filename=Political%20Guidelines%202024-2029_EN.pdf.

3 European Green Deal, https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en

4 Decision (EU) 2020/1782 of the European Parliament and of the Council of 25 November 2020 amending Decision No 573/2014/EU on enhanced cooperation between Public Employment Services (PES), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:32020D1782.

5 Council Recommendation on ensuring a fair transition towards climate neutrality, 2022, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=oj:JOC_2022_243_R_000.

6 The European Pillar of Social Rights Action Plan, https://op.europa.eu/webpub/empl/european-pillar-of-social-rights/en/.

7 Just Transition Mechanism, https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal/finance-and-green-deal/just-transition-mechanism_en.

8 Social Climate Fund, https://climate.ec.europa.eu/eu-action/eu-emissions-trading-system-eu-ets/social-climate-fund_en.

Plus[9], as well as the REPowerEU plan[10], the InvestEU Programme[11] and the Innovation Fund[12].

Skills are at the centre of the European Commission agenda. Roxana Mînzatu, European Commission's Executive Vice-President for Social Rights and Skills, Quality Jobs and Preparedness will lead the work on implementing the Union of Skills published in March 2025[13].



Who is the Toolkit for?

The Toolkit is targeted at PES staff in management positions at headquarters, at regional and local levels, as well as employment counsellors and staff working in the area of labour market information systems. It should also benefit other organisations working closely with PES on the green transition, including line ministries and other ministries, those in the training ecosystem, sector associations, social partners, non-governmental organisations (NGOs), local stakeholders.



How is the Toolkit organised?

The Toolkit is divided into five Sections:

1. **Section 1 presents PES strategies in relation to the green transition and definitions on green jobs.**
2. **Section 2 focuses on the identification of green skills.**
3. **Section 3 discusses the design of PES services for the labour force.**
4. **Section 4 covers PES activities for employers.**
5. **Section 5 provides some short conclusions.**

Sections 1 to 4 give a short background on key developments and challenges for the PES, showcasing inspiring practices put in place by PES in their own institutional and economic contexts. Each Section concludes with a checklist for PES on issues they may want to consider when developing strategies, producing labour market information, developing and implementing active labour market policies (ALMPs) and counselling services for the labour force and employers in relation to green skills and jobs.

The Toolkit is based on desk research in 2024[14], previous work for the European Network of Public Employment Services, inputs from the Taxonomy PES Network Working Group, a small taskforce of PES[15] which organised a call for practices from the PES Network. Wider examples beyond the EU are also provided, building on desk research[16]. Supporting the green transition is still a new field of activities for many PES, thus the evidence base is limited. Promising practices are taken as an illustration; however, they mostly lack thorough evaluations at the time of writing.

9 ESF+, https://european-social-fund-plus.ec.europa.eu/en.
10 RePowerEU Plan, https://commission.europa.eu/topics/energy/repowereu_en.
11 InvestEU Programme, https://investeu.europa.eu/investeu-programme_en.
12 Innovation Fund, https://climate.ec.europa.eu/eu-action/eu-funding-climate-action/innovation-fund_en.
13 For more information on the Union of Skills, https://commission.europa.eu/topics/eu-competitiveness/union-skills_en.
14 Please note some examples might be not running anymore but were relevant at the time of desk research.
15 The taskforce included the following PES: BE-ACTIRIS, DE, EL, ES, LV, LT, MT, SI.
16 World Association of Public Employment Services, PES and greening economies Catalogue of Actions, WAPES G7 Working Group Changing Labour Markets, 2023, https://wapes.org/blog/2023/07/12/catalogue-of-actions-new-wapes-g7-group-supports-greening-economies/?lang=en.

1 PES POSITIONING THEMSELVES IN THE GREEN TRANSITION

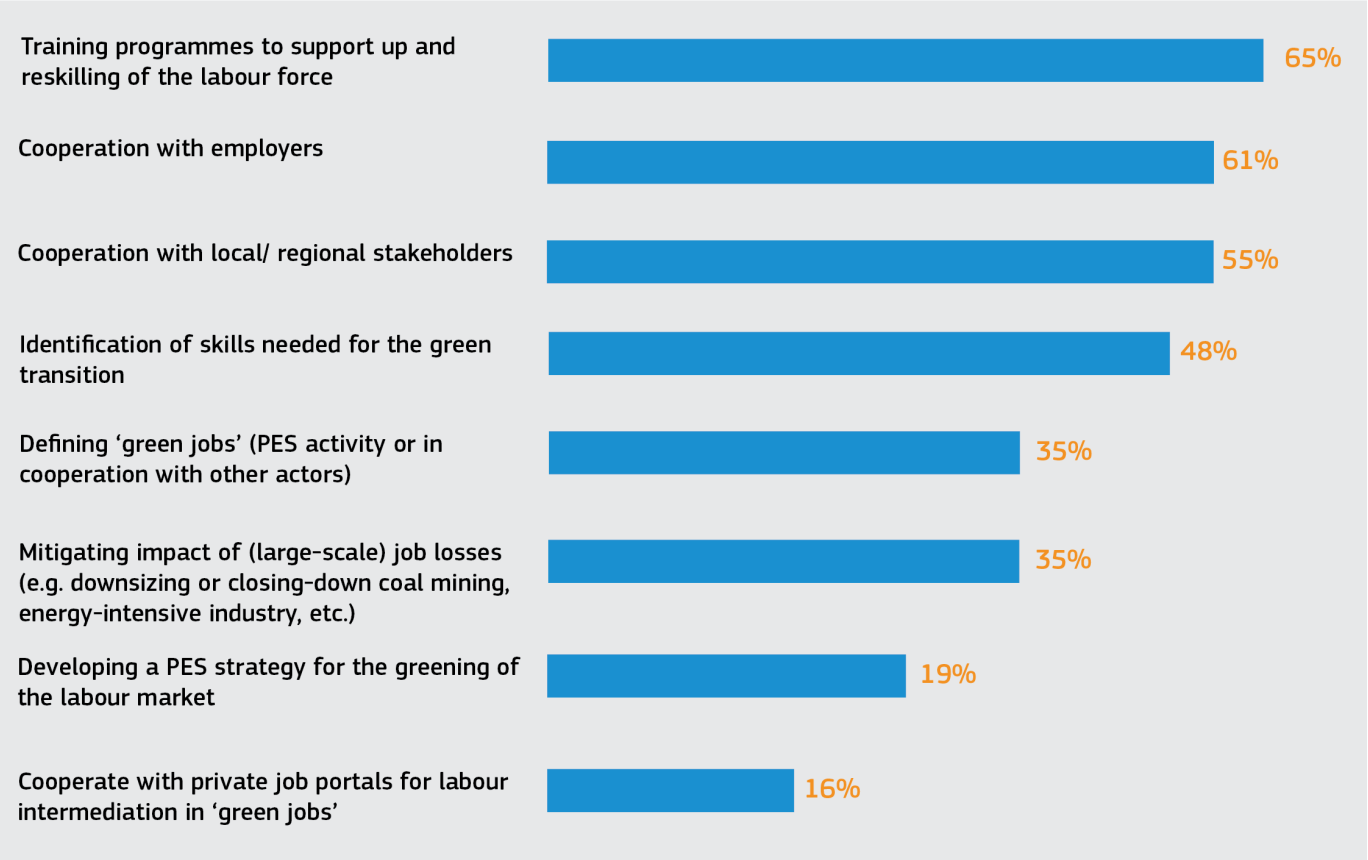
PES need to respond to the various impacts of the green transition on the labour market, which affect different PES functions (as outlined in Table 1).

Table 1. Impact of the green transition on PES services

PES service	Addressing the green transition in PES services
Labour market intelligence	Research (with partners) on the impact of the green transition, such as via the development of skills forecasting models.
Labour market taxonomies and classification systems	Defining what constitutes 'green' in occupations and skills in PES classification systems.
Support for employment adjustments in case of economic restructuring	Mitigating effects of the green transition linked to job losses, ensuring access to jobs by supporting job and geographical mobility and skills development.
Career counselling	Supporting jobseekers, including young people benefitting from dedicated programmes, to plan their career by providing information on present and future skills needs, including for the green transition.
Active Labour Market Policies (ALMPs) provision	Supporting jobseekers, including school leavers, to develop, adapt their skills, upskill, retrain and engage in the vocational training needed for the green transition. Supporting people at risk of losing their employment through early interventions, particularly in sectors affected by economic restructuring.
Integrating more vulnerable groups and individuals from disadvantaged backgrounds into the labour market	Providing access to adult learning, especially for vulnerable groups, focus on upskilling and reskilling, in the green economy. Provide support (through vocational guidance and counselling) to attract more women into green jobs, in particular in STEM fields related green jobs.
Support for employers	Supporting employers to better define their needs in their job offers. Developing skills/occupational profiles with employers. Supporting employers and employees to manage job-to-job transitions in sectors or regions affected by structural change.

As a result and to address sector-specific challenges, some PES have included the green transition in their strategies at national and/or regional level. In 2023, 6 of 31 PES (19% of respondents) had developed, or were in the process of developing, a strategy for the green transition of the labour market (France, Malta[17], Austria, Poland, Slovakia and Finland)[18]. Other PES initiatives are trigger by economic restructuring or come directly from national policies on greening of labour markets and related skills development. Several PES are involved in work to define and measure green jobs and identify related skills. Some countries are undertaking joint efforts to establish a common definition of green jobs.

Figure 1. Share of PES with strategies, measures, and activities to support the greening of the economy



Source: PES Capacity Questionnaire 2023, Part 1, Q5.5: What strategies, measures, and activities are in place at your PES specifically to support the greening of the economy?

17 The Maltese PES has implemented training initiatives which promote the upskill and reskill of the workforce vis-a-vis the green transition. There is however no official green transition strategy related to the labour market.

18 European Network of Public Employment Services, Trends in PES Assessment report on PES Capacity, 2023, https://employment-social-affairs.ec.europa.eu/news/pes-network-capacity-report-shows-latest-trends-public-employment-services-2024-05-06_en.

1.1 PES STRATEGIES TO ADDRESS LABOUR MARKET CHALLENGES LINKED TO THE GREEN TRANSITION

Strategies on greening the labour market help to secure political and operational commitment, ensuring that services are supported and adequately resourced. A strategic approach can be part of an overarching strategy of the PES or a self-contained strategy purely focusing on this topic.

Strategic approaches within PES

Strategic approaches within PES are often coherent with, or triggered by, wider initiatives, such as the United Nations (UN) Sustainable Development Goals (SDGs) or EU initiatives like the European Green Deal or the 2022 Council Recommendation on learning for the green transition and sustainable development[19]. Political or high-level support is often crucial to allocate resources for implementation and ensure commitment from decision makers. For example, the strategy of the **PES Forem** (Belgium Wallonia), has been influenced by national policymaking, **namely the priorities of the (previous) Walloon regional authorities**, and their action plan to give new impetus to vocational training (*Wallonie Compétences d'avenir*).

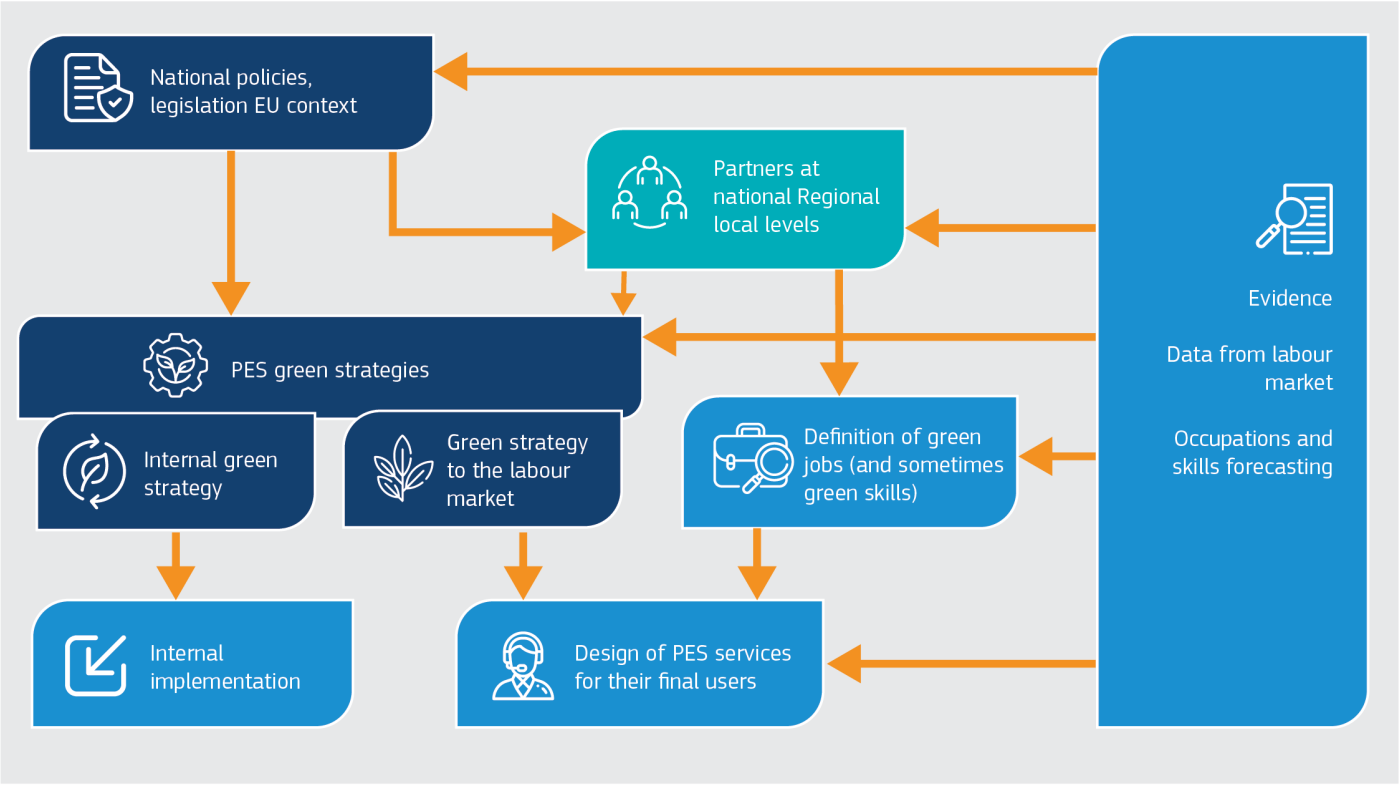
One of the six operational axes is 'Meeting the challenge of green transition', boosting the sustainable development dimension in training plans. The Regional Policy Statement (2019-2024) calls for a Climate Employment Renovation Alliance, with a focus on job creation in sustainable construction[20]. In turn, internal PES strategies on the green transition, as well as wider legislation and policies, influence the definition of green jobs used within PES (See Section 1.2)

Strategies need to be evidence-based and, ideally, backed up by **labour market intelligence, such as skills forecasting models** (see Section 2.2). However, most PES still face capacity issues in assessing the impacts of the green transition on jobs, and thus on the skills required, and in any event are not involved in forecasting[21].

Moreover, recent work by the PES Network stresses the need for more **partnership working** to capture trends in the green transition[22]. Stakeholders such as professional organisations, social partners, education and training sectors, national statistical offices, and institutions with specific expertise related to greening can provide expert input, e.g. on the changes of skills requirements of occupational profiles.

19 2022 Council Recommendation on learning for the green transition and sustainable development, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32022H0627%2801%29.
20 Regional Policy Statement 2019-2024, https://recherche.wallonie.be/en/home/a-propos-de-nous-1/declaration-de-politique-regionale.html.
21 The internal PES survey for the Taxonomy Working Group suggests that only seven PES (of 23 respondents) monitor the effects of the green transition in the labour market. Four PES have a dedicated team to work on the green transition. These are often quite small teams of experts providing analysis at national level and/or staff working on regional levels.
22 European Network of Public Employment Services, Greening of the labour market – impacts for the Public Employment Services, Small-scale study, 2021, https://op.europa.eu/en/publication-detail/-/publication/a5ce471b-f0dd-11eb-a71c-01aa75ed71a1/language-en.

Figure 2. Context of PES Strategies towards green labour markets



Source: own development.

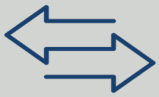
In terms of staff, few PES (including BE-Forem, DE, EL, PL)[23] have a dedicated team to work on the green transition. These are often relatively small teams of experts providing analysis at national level and/or staff working on regional levels. Some PES developed approaches to increase capacity of staff to provide PES services for the green transition, such as the examples below.

23 Internal PES survey for the Taxonomy Working Group, 2023.

Box 1. Staff capacity building for PES services for the green transition[24]



Formal qualifications: Bachelor study courses in Career Guidance and Labour Market Management at Germany's University of Applied Labour Studies (governed by the PES) include a module on occupational and labour science, which covers the evolving competence profiles due to ecological policies. A new module was introduced in 2023 on 'Sustainable Work and Labour Agencies – Future of Work and Integration in Times of Climate Change'[25].



Exchange and awareness-raising: In the French PES, staff are encouraged to consider ecologic change in strategic partnerships, for instance via sharing local examples to promote the green economy (in partnership with other actors, such as municipalities) on the intranet.



Staff training: Different formats of staff training can be applied. For example, the French PES provides one-hour online training on green and greening occupations for PES staff. Another example includes the delivery of 24 training modules for PES staff implementing the Intercept project, a project targeting young NEETs in Malta, Lithuania and Italy (see also Section 3.2)[26] . The modules cover labour market changes, work in the green economy and required skills.

Internal PES sustainability strategies

Some PES have developed internal sustainability strategies in line with activities of other public agencies to raise awareness of green issues among staff and clients, as an example. These strategies (which may be part of the broader PES strategy) aim at improving building efficiency, reducing waste or promoting sustainable mobility. For example, **PES Forem** (Belgium Wallonia) is focusing on internal aspects such as: limiting energy consumption and better energy management, sustainable transport for employees and users, a responsible public procurement policy, awareness-raising and training.

Another aspect of internal sustainability strategies is to update PES training equipment. In **Portugal,** Vocational Education and Training (VET) Centres (partly) managed by the PES were refurbished with technologically intensive equipment to accommodate training for professions such as wind and photovoltaic energy production.



TIP: the European Eco-Management and Audit Scheme (EMAS) supports organisations, including **public authorities**, to assess, manage and continuously improve their environmental performance.

24 World Association of Public Employment Services, PES and greening economies Catalogue of Actions, WAPES G7 Working Group Changing Labour Markets, 2023, https://wapes.org/blog/2023/07/12/catalogue-of-actions-new-wapes-g7-group-supports-greening-economies/?lang=en.

25 Ibid.

26 European Network of Public Employment Services, PES support to greening of the labour market, Thematic paper, 2023, https://op.europa.eu/en/publication-detail/-/publication/04ec652e-1ae8-11ee-806b-01aa75ed71a1.

To raise awareness internally, staff are also supported to meet organisational sustainability targets, as shown in the examples below.

Box 2. Staff awareness raising for internal sustainability aspects[27]



Green Teams and leadership: The internal strategy by PESGermany[28] around reducing emissions and sustainable procurement includes Green Teams in local PES who promote internal sustainability, as well as Green Leaders who showcase green issues in their teams.



Sustainability Champions: Outside of the EU, the UK Department of Work and Pensions (DWP) trains Work Coaches to raise awareness as Sustainability Champions, through collegial collaboration rather than written guidance.

Making use of new funding opportunities

National and EU funding has been used to support projects for the green transition. EU funds used include the Just Transition Fund and potentially the Recovery and Resilience Facility, as well as the European Social Fund+ that also allows for social innovation. However, a challenge in using it for green activities is to ensure continuity over time. For an ambitious green transition, this type of measure may need to become structural, i.e. innovative approaches and ALMP budgets may need to be taken over by mainstream ALMPs[29].

PES STRATEGIES: LESSONS AND CONSIDERATIONS FOR PES

Key takeaways



While more PES seem to focus on internal sustainability strategies, only few PES have developed strategies to facilitate the green transition[30], in line with national or/and EU level strategies, such as the Sustainable Development Goals, the European Green Deal or other EU-level initiatives. Strategies generally support PES approaches to defining, identifying green skills and jobs, as well as wider PES services.

27 World Association of Public Employment Services, PES and greening economies Catalogue of Actions, WAPES G7 Working Group Changing Labour Markets, 2023, https://wapes.org/blog/2023/07/12/catalogue-of-actions-new-wapes-g7-group-supports-greening-economies/?lang=en.

28 Operational implementation programme for ecological sustainability in the German Federal Employment Agency (Operatives ¬Umsetzungsprogram zur ökologischen Nachhaltigkeit in der Bundesagentur für Arbeit). 2024, https://www.arbeitsagentur.de/datei/operatives-umsetzungsprogramm-zur-oekologischen-nachhaltigkeit_ba048173.pdf.

29 World Association of Public Employment Services, PES and greening economies Catalogue of Actions, WAPES G7 Working Group Changing Labour Markets, 2023, https://wapes.org/blog/2023/07/12/catalogue-of-actions-new-wapes-g7-group-supports-greening-economies/?lang=en.

30 European Network of Public Employment Services, PES support to greening of the labour market, Thematic paper, 2023, https://op.europa.eu/en/publication-detail/-/publication/04ec652e-1ae8-11ee-806b-01aa75ed71a1.

Lessons learned, considerations and recommendations	Examples
The strategies for the green transitions (only few PES have strategies in place) should be coherent with those of other key stakeholders (such as other public bodies or social actors) at EU, national and local levels.	**PES BE (Forem)** focuses on job creation[31] insustainable construction, linked to the Regional PolicyStatement (2019-2024) that calls for a Climate Employment Renovation Alliance. **PES BE (Actiris) is focusing on specific sectors**, linked to the 'Shifting Economy'[32] strategy of the Brussels- Capital-Region's government.
Analyse the labour market impact of the green transition, such as type of jobs affected, characteristics of those employees, employers of jobseekers affected (e.g. age, education level). Such analysis can be applied to the economy as a whole or focus on a specific sector (e.g. jobs in construction) and/or geographical area.	**PES Poland** is involved in an ESF+ funded project 'Development and implementation of astrategic forecasting system for skills demand in the area of green and digital economy' to set up a forecasting system including on green skills (see Section 2.2 **Skills anticipation and informing on skills needs).**
Establish a stakeholder strategy with associations, skills councils, training institutions. Support local PES in exchanging and collaborating at local level with a variety of local stakeholders, such as municipalities, training institutions, employers and NGOs to work towards coherent approaches to support greening the local economy.	**PES Slovenia** emphasises partnership working in its 'Strategy on Greening of the Labour Market' in its Sustainability Report. **PES Norway** is running projects with universities to improve PES employees' knowledge of labour markets and their capacity to offer appropriate services and measures to their clients[33]. At **EU level**, high level strategic dialogues with key stakeholders address the green transition, such as in the automotive sector. See local collaboration approaches by various PES in 3.2 Targeted PES services and Active Labour Market Programmes (ALMP)

31 Forem, Anticipating skills and training needs in construction 2021, https://www.leforem.be/content/dam/leforem/fr/documents/chiffres-et-analyses/analyses/WCA-construction-analyse-VF-20211215.pdf.
32 https://shiftingeconomy.brussels/shiftingeconomy/.
33 Ibid.

Promote the green transition amongst staff and clients via PES' internal sustainability approaches.	Five PES (BE Actiris, BE VDAB, DK, FR, NL) set KPIs[34] in terms of internal sustainability, e.g. reducing energy consumption, waste management, sustainable mobility or use of renewable energy. PES France uses the international Assessing low-CarbonTransition[35] method, incorporating the Science based Targets Initative SBTi)[36] requirements.
Utilise EU funding to finance innovative aspects of the strategy. At the same time, ensuring sustainability of funding sources is crucial to implement strategies towards the green transition.	**PES Sweden** used EU funding to develop innovative recruitment (see Section 3.2). The PES worked on a mapping to define green skills via the Kickstart project, funded through React EU programme (p.20). In **Estonia** and **Poland**, the Just Transition Fund supports economic restructuring (see Section 3.2). Here the PES is involved in upskilling to support job-tojob transitions.

1.2 DEFINING GREEN JOBS

A definition of green jobs is needed for PES to produce reliable labour market intelligence and for the development of their services (for example, when granting employment subsidies for green jobs or jobs in green sectors). The need for a common definition is also outlined in the PES Network Memorandum to the next European leadership[37].

There is however no common definition of green jobs across the EU, which leads PES to innovate in this domain: while some PES have developed their own approaches, classifying green jobs is also not part of their core mission. Defining green jobs is also a complex issue for PES. As discussed by the PES Network Taxonomy Working Group[38], PES face several challenges in developing in-house definitions, largely because extensive data collection and multi-faceted expertise is needed. The capacity to define green jobs also depends on several factors, such as political emphasis, resources, or the

34 European Network of Public Employment Services, Trends in PES Assessment report on PES Capacity, 2023, https://employment-social-affairs.ec.europa.eu/news/pes-network-capacity-report-shows-latest-trends-public-employment-services-2024-05-06_en.

35 The Assessing low-Carbon Transition (ACT) initiative drives climate action by enabling benchmarking against advanced, science-based metrics. https://actinitiative.org/en/.

36 Science Based Targets, SBTi Corporate Net-Zero Standard, 2024, https://files.sciencebasedtargets.org/production/files/Net-Zero-Standard.pdf.

37 European Network of Public Employment Services, A PES Network Memorandum for the Next European Leadership - European Commission, 2024, https://employment-social-affairs.ec.europa.eu/news/pes-network-memorandum-next-european-leadership-2024-04-15_en.

38 European Network of Public Employment Services, Taxonomy Working Group, 2024, Internal report" Existing approaches on green jobs among PES", (forthcoming).

availability and resources of partners. In addition, recent work by the PES Network highlights the need to collaborate with employers and social partners[39].

Research, national statistical offices, and institutions with specific responsibility and expertise on greening provide expert input to labour market taxonomies, such as changes of skills requirements or occupational profiles.

Green jobs: existing definitions at international level

There is no universally accepted definition of green jobs, and definitions vary by country and depend on policy priorities or sector focus. The UN defines green jobs as "jobs in sectors that contribute substantially to preserving or restoring environmental quality and minimise waste creation and pollution"[40]. The ILO defines green jobs as "decent jobs that contribute to preserve or restore the environment, be they in traditional sectors such as manufacturing and construction, or in new, emerging green sectors such as renewable energy and energy efficiency"[41].

Approaches to defining green jobs

Depending on the Member State, different actors (e.g. a particular ministry or a governmental agency) may have adopted definitions, adding to the complexity of the landscape. Some PES have also engaged in defining green jobs for their own purposes.

In recent years, different definitions and measurement concepts have been developed by research. Two approaches are commonly used to define 'greenness': one focuses on the process involved, while the other looks at outcomes or results[42]:

- The process-based (or emission-based) approach determines the greenness of production based on the pollution it causes.
- The output-based approach determines the greenness of a product based on its potential beneficial effects on the environment.

However, these two approaches do not fully consider the entire workforce and economy. Table 2 summarises information on approaches to defining green jobs that go beyond output and process.

39 Ibid.
40 OECD, Job Creation and Local Economic Development 2023: Bridging the Great Green Divide, OECD Publishing, Paris, 2023, https://www.oecd.org/en/publications/job-creation-and-local-economic-development-2023_21db61c1-en.html.
41 Green jobs - ILO Research Guides at International Labour Organization, https://libguides.ilo.org/green-jobs-en.
42 Bontandini, F. and Vona, F., Anatomy of Green Specialisation: evidence from EU production data, 1995-2015, OFCE working paper 21/2020, https://link.springer.com/article/10.1007/s10640-023-00781-7.

Table 2. Approaches to defining green jobs

Approach	Information
Sector-based approach	Considers jobs to be green if they are found in sectors whose outputs or production processes substantially contribute to preserving or restoring environmental quality. *Advantage: It may be easier to target employment incentives in green sectors in terms of programme implementation.*
Occupation-based approach	Defines green jobs as having shared commonpurposes and requirements, i.e. green jobs contribute to safeguarding environmental quality in various ways. The focus is removed from the greenness of either the process or the output and turns instead to what is required (tasks, skills, knowledge) for a specific occupation to be considered green. *Advantage: Occupation-based definitions may be valuable for vocational guidance and providing information on occupations.*
Task-based approach	A more nuanced understanding of 'greenness' within occupations, capturing the extent to which workers engage in green tasks. Tasks refer to what workers are expected to do in the workplace: they are the unit of work that produce an output (demand side), while skills are the capability to perform various tasks (supply side). *Further information on the advantages is included below.*

Sources: European PES Network, Taxonomy Working Group (forthcoming For green-driven occupations; see OECD, Employment Outlook 2024).

Focus on the task-based approach

Few PES are looking at the task-based approach (see Annex 1), whether implicitly or explicitly. The task-based approach allows PES to observe the evolution of what workers do, both in existing and new occupations, helping to identify the skills of increasing relevance for the green transition. It can be adapted to different definitions of green and applied to different datasets at European, national and local level.

Existing definitions of green skills and tasks (ESCO green skillset (see Section 2), or the United States' **Occupational Information Network database** (O*NET)

definition of green jobs and tasks) have served as an inspiration and benchmark for PES with more established definitions on green jobs, such as PES in France and Germany. O*NET contains detailed occupation-level information on the tasks and skills involved in jobs, as well as a list of tasks and jobs labelled as green. It also provides information on whether these occupations require enhanced, new, or emerging green skills. By examining the tasks associated with green occupations and industries, the database provides empirical evidence on the drivers and effects of green employment. The limitations related to using O*NET include that the findings relate to the US market, and it is unclear at the time of writing to what extent they can be applied to the EU labour market.

Through the O*NET's task information, researchers[43] constructed a measure of greenness, with an emphasis on occupations with a larger share of green tasks: 'The greenness indicator is defined as the ratio between the number of green-specific tasks and the total number of specific tasks done in one occupation'.

Piloting a common approach: PES Network Taxonomy Working Group

The PES Network Taxonomy Working Group is working on a common approach to defining green jobs as part of its work on green jobs and skills taxonomies. It aims to pilot this common approach with both European and national taxonomies[44].

At EU level, the task-based approach will be tested on ESCO data. At national level, PES are invited to pilot the same methodology using their national classifications. The goal is to compare the definition of occupational greenness from ESCO with a definition using different textual corpus.

DEFINING GREEN JOBS: LESSONS AND CONSIDERATIONS FOR PES

Key takeaways



Defining what constitutes 'green' helps PES to better target their interventions. However, definitions of green jobs vary between stakeholders and are based on different methods primarily sector-based, occupation-based, or task-based. The task-based approach has the advantage of looking at tasks and skills at the same time, which informs evolving skills needs across occupations, in line with the 'skills-based approach' and the notion of skills portability between jobs.

43 Vona, F., Marin, G., Consoli, D. and Popp, D., Environmental Regulation and Green Skills: an empirical exploration, Journal of the Association of Environmental and Resource Economists, Vol. 5, Issue 4, 2018, pp. 713-753; Vona, F., Marin, G. and Consoli, D., 'Measures, drivers and effects of green employment: evidence from US local labour markets, 2006–2014', Journal of Economic Geography, Vol. 19, Issue 5, 2019, pp. 1021-1048.

44 Seven out of 23 PES in the Taxonomy Working Group.

Lessons learned, considerations and recommendations	Examples
Collaborating with partners and ensuring coherence with definitions of green jobs and skills used by other national stakeholders, such as ministries.	**PES France** uses the definition by France's National Observatory for Jobs and Occupations of the Green Economy (Observatoire national des emplois etmétiers de l'économie verte, Onemev).
If the PES has no definition and/or list of green skills, exploring the use of the green skillset provided by ESCO, building on outcomes of the PES Network Taxonomy Working Group's work on green taxonomies.	See Annex 1 and examples in Box 3.
Depending on internal capacity, explore innovative methods, such as big data analysis, to identify green jobs and related skills.	Artificial intelligence (AI) to analyse online vacancy data on green skills (e.g. French PES, see Annex 1, or. Finding jobs in green industries: methodology, developed by Nesta, **German PES**' text-mining approach to identifying green skills).
Cooperation and coordination with other stakeholders (e.g. employers and training providers) on 'green' economic activities, jobs and occupations to ensure exchange on what is considered green and to gain buy in.	**PES Croatia** is part of a working group to review green skills sets, **PES Finland** collects information more informally, **PES Forem** (Belgium Wallonia) interviews stakeholders on the impact on occupations and skills.

2 IDENTIFICATION OF GREEN SKILLS

Green skills are the knowledge, abilities, values and attitudes needed to live in, develop and support a society which reduces the impact of human activity on the environment'[45].

A list of "green skills" can be found in the European Classification of Occupations, Skills, and Competences (ESCO)[46] which has adopted the definition of green skills developed by Cedefop. From a sector perspective, Eurostat is also defining and measuring the environmental goods and services sector (EGSS[47]) which provides a framework to collect data on the environmental sector.

45 Cedefop, Green skills and environmental awareness in vocational education and training: Synthesis report, Research Paper No 24, 2012, https://www.cedefop.europa.eu/en/publications/5524.

46 ESCO identifies knowledge and competences in over 3 000 occupations.

47 https://ec.europa.eu/eurostat/cache/metadata/de/env_egs_esms.htm.

Using ESCO to define green skills

The **ESCO green skillset** may provide a starting point for PES to define green jobs and skills. While ESCO is not yet directly used by all PES, the European standard terminology is implement-ed by PES in the context of the EURES regulation's implementation, suggesting the usefulness of working towards a common task-based definition based on ESCO data. Several PES review existing nationally classified skills against ESCO's green and digital skills or adopt ESCOs green skills if no national definition exists.

Box 3. Use of ESCO among PES

The European Commission has published a taxonomy of skills for the green transition in ESCO. It includes 591 skills and knowledge concepts considered relevant for a greener labour market. Examples of green skills include how to conduct energy audits, measure the sustainability of tourism activities, and train staff on recycling programmes. Examples of green knowledge concepts include emission standards and ecological principles. Green transversal skills include evaluation of the environmental impact of personal behaviour and adoption of ways to boost biodiversity and animal welfare.

The **Swedish** PES is working on a taxonomy for green jobs. As part of the Kickstart ReactEU programme funded by the ESF, the PES is mapping ESCO-defined green and digital jobs and translating them into its own definitions. The aim is to create a unified picture of green and digital skills and make ESCO-defined skills compatible with Swedish labour market skills definitions.

PES in **Portugal** and **Greece** define those jobs requiring a certain amount or specific type of green skills as green jobs. **PES Croatia** participates in regular meetings and working groups on green skills and subsidised employment in green occupations and sectors (ALMP Working Group; Sectoral Working Group). Green skills are identified via an initial mapping of skills from the national qualification framework to compile a catalogue of skills necessary for the green transformation/transition of the economy. This catalogue is used to identify green skills and support education and training measures through vouchers. The catalogue is continuously supplemented with green and digital skills newly added to the national qualification framework and ESCO.

The green transition is expected to have a major impact on skills requirements and occupations. This will create the necessity to invest in training of the workforce, for example to employ candidates with expertise in working with battery cells, heat pumps, solar photovoltaics and wind energy. It should be noted in this context that, although the shift to a net-zero economy may cause job losses in male dominated sectors, overall projected job growth key sectors is likely to benefit men more than women[48]. As outlined in previous work by the PES Network, PES must be aware that the skills needs emerging from the green transition will impact on occupations in various ways:

48 International Monetary Fund, Green Jobs and the Future of Work for Women and Men, Staff Discussion Notes 2024, 003 (2024), https://doi.org/10.5089/9798400286049.006.

- Expansion of occupations demanding skill adjustments. There is a distinction between occupations requiring minimal skill adaptation within existing roles and those necessitating significant skill adjustments to align with environmentally sustainable practices. This is in line with the notion of green and 'greening' jobs;
- Growth of entirely new occupations in the green economy; and
- Decline in traditional occupations that become obsolete due to the twin transitions.

2.1 IDENTIFYING DIFFERENT TYPES OF GREEN SKILLS FOR THE GREEN TRANSITION

Identifying green skills is a complex task, undertaken by only a minority of PES[49]. PES need to identify new and transferable skills that can support the transition into jobs with an increasing workforce demand, but it is often difficult to grasp the new skills required and those that can be transferred between occupations or sectors[50]. The 'greening of jobs and skills' is a continuum process, involving skill adaptation, upskilling and reskilling.

Skills needed for the green transition include new technical skills linked to new energy technologies (e.g. designing water conservation systems), digital and data analytical skills (e.g. managing use of resources efficiently), and skills to implement the principles of a circular economy (e.g. sustainability of products and processes, reuse of materials).

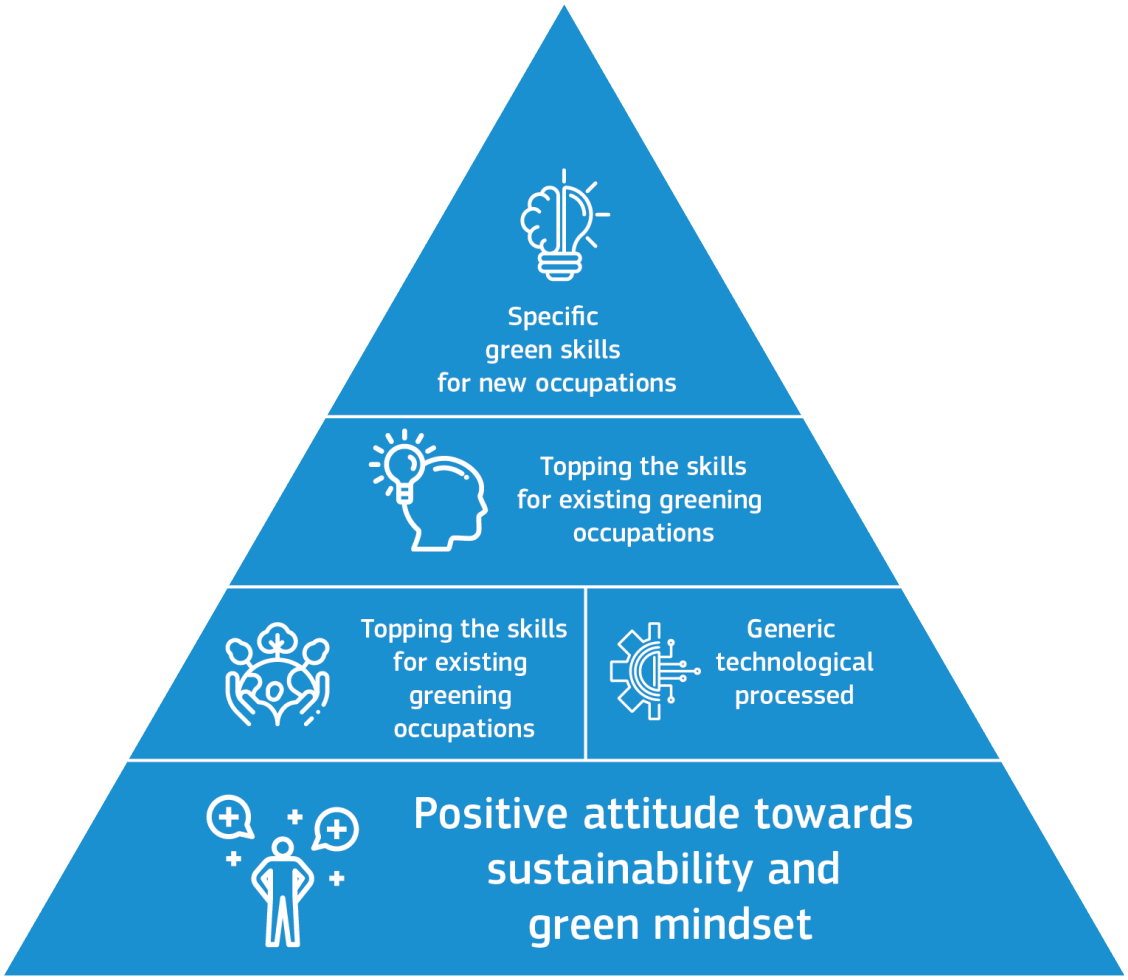
Box 4. Main areas of green skills and competences: a definition

The European Commission and CEDEFOP have identified the **following five main areas as environmental skills and competences**[51]:

- Evaluate environmental impact of personal behaviour
- Adopt ways to reduce the negative impact of consumption
- Adopt ways to reduce air, noise, light, water or environmental pollution
- Adopt ways to foster biodiversity and animal welfare
- Engage others in environmentally friendly behaviours

49 Nine of the 24 PES identify green skills (Taxonomy Working Group).
50 European Network of Public Employment Services, PES support to greening of the labour market, Thematic paper, 2023, https://op.europa.eu/en/publication-detail/-/publication/04ec652e-1ae8-11ee-806b-01aa75ed71a1.
51 European Commission, Towards a structured and consistent terminology on transversal skills and competences | European Skills, Competences, Qualifications and Occupations (ESCO), 2022, https://esco.ec.europa.eu/en/about-esco/publications/publication/towards-structured-and-consistent-terminology-transversal.

Figure 3. Typology of green skills



Source: Pavlova, 2017

Green skills encompass not only technical skills and soft skills, but also attitudes, behaviours and values.

Green skills enhance existing skills for greening all occupations and the acquisition of specific green skills for new green jobs in industries such as waste and water management, renewable energy, eco-tourism, and environmental consulting services (see Figure 3). This indicates that while **generic green skills** and skill enhancements are necessary for all occupations to transition to greener practices, new green occupations demand additional **specialised green skills**.

Some of these skills are portable from one sector and occupation to another. The International Labour Organization (ILO) has identified a number of **portable skills** to train environmental awareness[52], including strategic and leadership skills, willingness to learn about sustainable development, management and business skills, as well as entrepreneurial skills, consultation skills to advise consumers, systems and risk analysis, language skills and information technology (IT) skills.

52 Strietska-Ilina, O., Green skills for green jobs, Presentation, ILO, 2015.

Portable skills also include technical skills (e.g. in civil engineering) that can be used for different kind of energy sources. More specifically, transversal green skills and competences can be classified into[53]:

- **Cognitive competencies:** Environmental awareness and a willingness to learn about sustainable development, systems and risk analysis, ability to assess, interpret and understand the need for change and the measures required, ability to identify opportunities and create new strategies to respond to green challenges.
- **Interpersonal competencies**: Coordination, management and business skills to facilitate holistic and interdisciplinary approaches that encompass economic, social and ecological objectives, communication and negotiation skills to discuss conflicting interests in complex contexts, marketing skills to promote greener products and services.
- **Intrapersonal competencies:** Adaptability and transferable skills that help workers to learn and apply new technologies and processes required to green their jobs, and entrepreneurial skills to take the opportunities presented by low-carbon technologies.
- **Technological competencies**[54]**:** Quantification and monitoring (waste, energy, water), management systems (waste, energy, water), procurement and selection, material use and impact quantification, impact and use minimisation, impact assessment, risk management.

The green transition emphasises the need for training and skill transferability, with generic skills being transferable to many similar tasks (to varying degrees)[55]. Some PES are starting to look at how workers in traditional sectors can be upskilled to work in green occupations, building on sector-related insights necessary to identify the specific skills needed for the green transition and portability of skills (e.g. **Germany** and **Sweden**). Table 3 presents an example from Spain of green competency requirements adding to the skillset required in certain traditional sectors.

53 Green Skills Resources, based on Pavlova, 2014, and a study of per capita, https://www.greenskillsresources.com/category/generic-green-skills.

54 There is some debate about whether technological competencies can be included in the category of generic skills; however, it is justified if transferable elements of the technical skills are considered.

55 Rodrigues, M, Fernández-Macías, E. and Sostero, M., A unified conceptual framework of tasks, skills and competences, Seville, European Commission, 2021, JRC121897.

Table 3. Green activities in traditional sectors: an example from Spain

Traditional sector	Green activity
Auxiliary automotive industry	Components for wind turbines
Electronic components and electricians	Components for wind turbines
Civil public works	Construction of thermoelectric generators
Chemical and electronic industries	Photovoltaics
Agriculture	Biomass activities
Shipyards	Offshore wind farms
Plumbers	Solar thermal energy

Example for potential skills transferability. Source: CEDEFOP (2018); Skills for green jobs in Spain; an update.

Working closely with partners to identify green skills

The identification of skills often requires extensive external expertise, and many PES may profit from expertise by partners. For example, the Estonian PES is cooperating with the Ministry of Education and Research to develop green skills curricula for higher education, micro-credentials, and training. It also cooperates with the Estonian Qualifications Authority, which researches future skill needs (including green skills), and with a data science partner to develop a skills gap analysis tool (mapping green skills against the labour market demand).

Some PES work locally to gather knowledge on required green skills from local partners. For example, the **Slovak PES** has consulted representatives of individual sectors and industries, while **Finland's l**abour market training experts in ELY (Centre for Economic Development, Transport and the Environment) Centres identify relevant green skills needs.

Restructuring in specific sectors, where industries enter a process of environmental transition, affects often whole regions. PES or other national agencies analyse relevant sectors impacted by the green transition. Poland's Ministry of Family, Labour and Social Policy undertakes expert surveys on the possible impact of the energy transition on relevant sectors (e.g. mining), while France has indepth qualitative studies on occupations in sectors affected by the green transition.

Example: France – competences required for the circular economy

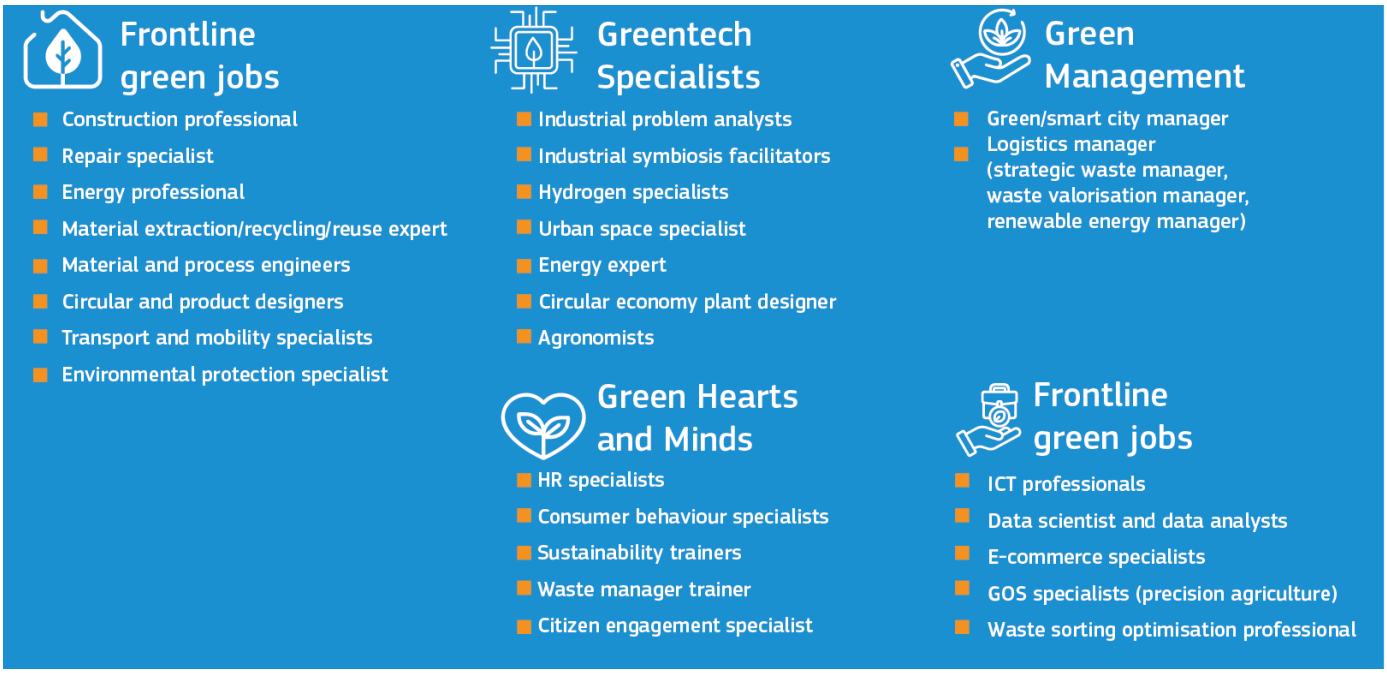
In France, the Ministry for Ecological Transition has identified **competences needed for the circular economy**[56]. The applied eco-concept differentiates between competences, such as analytical competences, use of technologies and problem-solving skills applied to the production and distribution of goods, and knowledge of materials, physics, chemicals, production processes, regulatory issues, specific norms, public tenders, etc. Specialists in eco-conceptions are mainly demanded in combination with other skills and competences in the small and medium-sized enterprises (SME) sector.

56 Ministère de la transition écologique, Compétences pour l'économie circulaire, 2020.

2.2 SKILLS ANTICIPATION AND INFORMING ON SKILLS NEEDS

Both sustainability and digitalisation are reshaping the labour market, as highlighted by CEDEFOP's Skills Forecast. High-skill, so-called, 'thyroid occupations'[57] (e.g. waste treatment and alternative fuels engineers) are driving innovation, while medium-skill technical roles (e.g. biogas technicians) remain crucial. Information and communications technology (ICT)-related jobs (e.g. data scientists) will also support the green transition. Automation may shift roles from production to office-based work. Managerial positions will guide businesses towards sustainable practices, while human resources (HR) officers will ease worker transitions and address skill shortages.

Figure 4. Key occupations for the green transitions (selected sectors, experts' assessment)



Source: Cedefop Skills forecast in cities, waste management, agri-food and circular economy

According to the European Commission, if implemented properly, the green transition could create around one million additional quality jobs in the EU by 2030, and two million by 2050, in sectors such as construction, ICT or renewable energy[58], circular economy, sustainable transport[59]. While these sectors are likely to face skills shortages, the impact of the green transition could also lead to job losses, for example in some mining activities or fossil-fuel based energy production, as well as in parts of the automotive sector. A number of studies aim to project future workforce needs.

Forecasts are not directly comparable, as they make different assumptions based on the type of policies implemented.

57 Cedefop names "thyroid occupations" as high- and middle-skilled occupations that invent, design, and concep-tualise green transition solutions (such as engineers, process designers, and scientists – research and develop-ment) and implement it (such as managers, lawyers, renewable energy technicians, and communication spe-cialists). https://www.cedefop.europa.eu/en/news/circular-economy-step-skill-needs-and-importance-thyroid-occupations.

58 European Commission, Stepping up Europe's 2030 climate ambition - Investing in a climate neutral future for the benefit of our people, SWD(2020) 176 final, https://eur-lex.europa.eu/legal-content/en/ALL/?uri=CELEX:52020DC0562.

59 Ibid.

Most forecasts predict a small net job creation, with variations across different sectors and regions in Europe, linked to regional disparities in economic structure. Some forecast examples are presented in Table 4 below:

Table 4. Estimates on future skills needs for the green transition and employment impact

Source	Method	Employment Impact	Sector Impact	Skills Impact
European Green Deal skills forecast scenario Cedefop (2021)[60]	Skills forecast scenario until 2030. Assumptions based on various Commission communications on the sectors most targeted by the European Green Deal, from carbon emissions reduction and sustainable economy policies[61]	1.2% additional growth, 2.5 million jobs by 2030	Gains and losses across sectors, redirection towards cleaner production	Employment gains are slightly more pronounced for some of the medium- and low-skilled occupations (highest for crafts and related trade workers (+1.7%) and plant and machine operators and assemblers (+1.6%) and in elementary occupations (+1.4%). Among high skilled occupations, employment is set to increase most for legislators, senior officials and managers (+1.4%). Technicians and associate professionals (+1.3%) and clerks (+1.2%) follow closely.

60 Cedefop, The green employment and skills transformation: insights from a European Green Deal skills forecast scenario, 2021. https://www.cedefop.europa.eu/files/4206_en.pdf.

61 Scenario assumptions relate to the achievement of the 50-55% emission reduction target by 2030, sector- or technology-specific policies contributing to the emission reduction target, and circular economy plans. The assumptions incorporate increased energy efficiency, and the cost needed to achieve it – linked to the Renovation wave, changes in the EU emissions trading system aimed at accelerating the renewable energy transition in road transport, and policies encouraging less carbon-intensive technology. The impacts of future price increases of polluting energy sources (through taxation) on road transport and buildings are also incorporated. Assumptions relating to the circular economy plan include increased recycling rates in sectors with high resource usage (e.g. electronics, plastics, vehicles) and harmonisation of the waste collection and management systems across the EU.

Fit for 55 climate package: Fit for 55 climate package: Impact on EU employment by 2030 Eurofound (2023)[62]	Projections from a global macroeconomic model how the Fit for 55 policy package affect the territorial, sectoral and occupational structure by 2030	Net creation of 204 000 jobs	Higher impact in mining/extractive regions (e.g., Poland, Romania), benefits construction sector	Boost in mid-paid and mid-low paid jobs, no tertiary qualifications required
OECD Skills Outlook (2023)[63]	Projections for the Fit for 55 package, modelling analysis comparing different scenarios	Job creation in high-skill occupations, declining opportunities for blue-collar workers	A switch from fuel-based energy towards renewable energy and a structural reallocation towards the service sectors. Together, these effects result in changes in employment by occupational category, in particular, towards more labour-intensive sectors.	Growth in demand for: interacting with computers, thinking creatively, analysing data, communicating externally.

An analysis of recent trends in the composition of green jobs and green-driven jobs by OECD (2024)[64], suggests that the net-zero transition is creating and will continue to create jobs that tend to be concentrated in high-skill occupations and represent an opportunity mainly for highly educated, urban workers who have the competences required for these jobs. In contrast, for low-skilled workers, green-driven occupations may not be a sufficiently attractive alternative to jobs in the rest of the economy (including high-emission industries). Several countries are developing skills anticipation tools, such as Poland and Belgium (Wallonia region), see below:

Examples: Skills forecasting project of the Polish PES and occupations of the future in Belgium-Wallonia

Co-financed by ESF+ 2021-2027, the main goal of 'Development and implementation of a strategic forecasting system for skills demand in the area of green and digital economy' is to im-prove the supply of skills demanded in the digital

62 Eurofound, Fit for 55 climate package: Impact on EU employment by 2030, 2023, https://www.eurofound.europa.eu/en/publications/2023/fit-55-climate-package-impact-eu-employment-2030.
63 OECD Skills Outlook 2023: Skills for a Resilient Green and Digital Transition, OECD Publishing, Paris, https://www.oecd.org/en/publications/oecd-skills-outlook-2023_27452f29-en.htm.
64 OECD, Job Creation and Local Economic Development 2023: Bridging the Great Green Divide, 2023, https://www.oecd.org/en/publications/job-creation-and-local-economic-development-2023_21db61c1-en.html.

and green economy in **Poland** by 2050.

The forecasting will integrate data on green and digital skills from existing systems, as well as incorporating a function generating forecast. The generated forecasts will then be used to formulate recommendations for activities and policies to develop skills to meet the needs of the green transformation. They will also be made more widely available in annual reports.

The PES **Forem (Belgium Wallonia) has launched a project to identify the occupations of the future for Wallonia over the next five years.** It aims to adapt and improve the range of services (whether organised by the PES or another player in the market). Forem's Market Analysis and Training Department (AMEF), after consulting more than 300 Walloon experts, published an initial analysis of the occupations of the future. This wide-ranging study took a broad look at the various sectoral development factors and their presumed effects on the development of occupations.

Source: Taxonomy Working Groups answers to the PES Network survey.



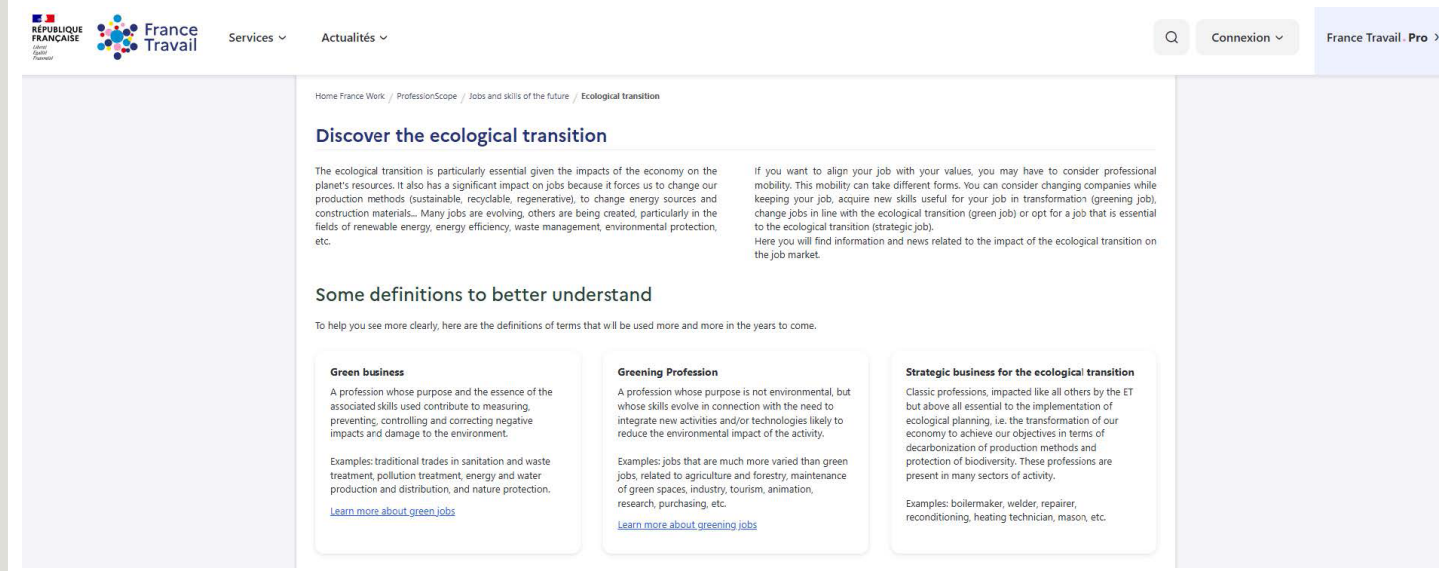
Using big data analysis to identify green skills: PES France uses text analysis to identify green skills. On EU level, Cedefop and Eurostat set up a database on online job advertisement to map how the greening of economies in Europe impacts labour markets. This data can be used to monitor skills need at sectoral or regional level, but it can also update on definitions on green skills and occupations.

Some PES have introduced green jobs and skills in their career guidance and vocational guidance information (e.g. Green section in France's *MétierScope*, see example below, and in Germany's **BERUFENET**[65]).

PES are key in managing employment impacts of transformation processes. While the complexity of defining green may limit the capacity of PES to provide advice on upcoming green employment opportunities, some PES are increasingly integrating green occupations and required skills in their labour market information system to inform training providers, jobseekers, employers, and local, regional and national stakeholders in the field of economic development and employment.

65 https://web.arbeitsagentur.de/berufenet/.

Example: Career counselling site by French PES



Source: France Travail, https://candidat.francetravail.fr/metierscope/metiers-avenir/transition-ecologique[66]

Gender imbalances in the green transition need to be more considered by PES, as job growth is expected in sectors in which women are underrepresented. For instance, EU-level projects such as the Women4Green project[67] raise awareness of STEM subjects amongst women and girls.

IDENTIFYING GREEN SKILLS: LESSONS AND CONSIDERATIONS FOR PES

Key takeaways



The anticipation of green skills allows PES to provide high quality labour market information and to plan PES service delivery more strategically. Research shows that the green transition is likely to continue to create new jobs, but job losses are expected in some sectors. Hence, knowledge about the different types of skills required for the green transition helps PES to develop services and to inform customers. Notably, the portability of skills is challenging to quantify. In order to mitigate those knowledge gaps, PES can work with data to anticipate skills or in partnership with experts to gather input locally.

66 MétierScope, https://candidat.francetravail.fr/metierscope/
67 https://women4green.eu/project/.

Lessons learned, considerations and recommendations	Examples
Utilising research on skills needs and portability of skills in different sectors and occupations in the course of the green transition, or developing national models. This also includes understanding differences of various forecasts which work with different assumptions.	Cedefop skills forecast scenario, as well as labour market skills forecasting by other institutions or organisations such as the European Commission, Eurofound, OECD and ILO (see Table 4.).
Analysing online job advertisements and other labour market data (such as those related to training or qualifications) can add to valuable insights for analysing the development of skills demanded.	PES **France** example in Annex 1, as well as work done by Cedefop
Using sector-related insights by partners is necessary to identify specific skills needed for the green transition and reflect on the portability and transferability of such skills.	Examples of PES **Croatia** in Annex 1 and PES **Germany** and PES **Sweden** in Section 3.2.
Supporting counsellors in informing PES clients about future skills needs.	See **France** and **Germany** examples in Section 2.2, use of leaflets, campaigns on green jobs and education.

3 DESIGNING PES SERVICES AND MEASURES FOR THE LABOUR FORCE

In addition to providing jobseekers with information and career guidance, PES have several labour market programmes[68] in place to support the workforce to cope with the green transition. The main measures include training, employment incentives, direct job creation programme, and financial support through unemployment benefits and transition benefits.

As the main challenges of the green transition consist of upskilling and reskilling, as well as identifying transferable skills for job matching, the key activities of PES need to focus on skills. PES have started to facilitate training for the green transition (both focused on technical and soft skills), although determining training content still needs further work[69].

3.1 TRAINING FOR THE GREEN TRANSITION

Learning systems are crucial for ensuring that workers stay employable and productive throughout their careers, as skill requirements evolve. A 'skills- based approach', so a focus on continuous learning (rather than primarily qualifications), helps to facilitate job-to-job transitions via portable skills and upskilling for specific green occupations.

PES' involvement in steering training offers towards skills needed for the green transition depends on their mandate and how they refer their clients to training measures, in a context where a large

68 Labour market programmes database, https://ec.europa.eu/social/main.jsp?catId=1143&intPageId=3227&langId=en.
69 Ibid.

majority of PES are responsible for upskilling and reskilling, but most of the time in cooperation with partners[70]. Some PES cooperate with training providers and steer training content via training vouchers and contracting systems. Other PES provide training themselves. Table 5 below describes some of the training initiatives undertaken by PES.

Of course, the role of employers and professional sectors is crucial in training dynamics. A recent report[71] underlines that training (and education) in some sectors that are key to the green transition (e.g the construction sector) is lagging behind.

Table 5. Training approaches for the green transition

Country	Training content	Target group	Further information (delivery models)
Austria	VET innovative training project for alternative energies and energy-efficient use of energy. Individual training plans are drawn up Environmental Foundation Programme's workplace-based training	Students at vocational schools and colleges, apprentices, jobseekers, employers, individual training for employees, 50% of places are reserved for women	Climate Protection Competence Centre, a joint measure with the PES and Vocational Training Institute (BFI), as well as in cooperation with employers It provides training in the areas of waste and resource management, green mobility, energy and building technology, agriculture and forestry. Research, education and counselling are also offered.
Belgium, Wallonia, (Forem)	Training centres and centres of excellence – some specialised in construction and environment (Construforms, Wallonie-Bois, Environnement and the Namur - BW construction training centres)	People in employment and jobseekers, employers	23 training centres, run by the PES for in-demand and future-oriented competences in areas with strong economic development potential[72]
Croatia	Accredited providers provide training (microcredentials, partial and full qualifications) based on green skills identified (see page 20)	People in employment and jobseekers	Voucher scheme for upskilling run by the PES

70 European Network of Public Employment Services, Trends in PES Assessment report on PES Capacity, 2023, https://employment-social-affairs.ec.europa.eu/news/pes-network-capacity-report-shows-latest-trends-public-employment-services-2024-05-06_en.

71 Fulvimari A., Garaffa R., Kunertova L., Van Der Vorst C. and Weitzel M., Labour Market transitions and skills investment needs in the green transition - a new approach, Fair Transition Working Paper series, European Commission, 2025.

72 https://www.leforem.be/centres-de-competence.html.

Finland	Labour market training includes solar panel installer, photovoltaic installer, circular economy expert, low-carbon expert, climate change expert, specialist qualification in the environment sector, business security expert	Jobseekers	PES (ELY Centres) plan and procure training
France	10 000 training courses[73] in ecological transition jobs with a focus on construction, public works, logistics, waste recovery, recycling	Young people and jobseekers	Training courses are co-funded by the French PES
Greece	Training in smart homes and cities, green entrereneurship, ecological design, bioclimatic design, use of new technologies, renewable energy, electric mobility	Jobseekers	Vocational Schools of Apprenticeship run by the PES
Portugal	Training courses in wind and photovoltaic energy production, hydrogen production, energy efficiency, electric mobility, water efficiency, forest management	Jobseekers and people in employment	Recovery and Resilience Plan (RRP) investment into 55 vocational training centres, 31 directly managed and 24 partly managed by the PES
Portugal	Green Skills and Jobs programme focuses on energy efficiency, renewable energy, water efficiency, sustainable mobility, circular economy	Workers in companies directly or indirectly affected by the increase in energy costs and jobseekers	EUR 20 million allocated to training providers and employers

Attracting workers and adapting their skills to new demand in the labour market can be a challenge in regions where new jobs are created in high volumes. One prominent example is the north of Sweden that has a good supply of renewable energy and access to raw materials and natural resources. The Swedish PES have set up a regional office for this transition (see also Section 3.2)[74].

Evaluations of ALMPs show that skill-related training and occupational re-orientation programmes can be effective in increasing re-employment probabilities, particularly in the medium and long term. They have also been shown to have some effect on re-employment wages after displacement[75].

73 Ministry for Ecological Transition and Solidarity and Ministry of Labour (Ministère d'Etat, Ministère de la Transition écologique et solidaire, Ministère du travail), Press release, 14 May 2018, Lancement des 10 000 formations aux métiers verts et verdissants dans le cadre du Plan d'Investissement dans les compétences - France Travail | francetravail.org.

74 Arbeitsförmedlingen, Tackling the labour and skills shortages linked to the green transition in the North of Sweden. Presentation by Shawn Mendes, Director of International Affairs.

75 Eppel, R., Huemer, U., Mahringer, H., & Schmoigl, L. Evaluierung der Effektivität und Effizienz von Qualifizierungsförderungen des Arbeitsmarktservice Österreich. WIFO. 2022.

3.2 TARGETED PES SERVICES AND ACTIVE LABOUR MARKET PROGRAMMES (ALMPS)

PES offer services to a wide range of participants, including workers at risk of becoming unemployed as a result of the green transition, or those in need of skills adaptation, employees, NEETs, and unemployed people.

Target Group	PES services and ALMPs	Tools
Workers at risk of becoming unemployed	Identifying sectors and workers affected Managing career transitions via skills assessment, upskilling (focus on specialised green skills), making sure that jobseekers can access other/new jobs (requiring occupational or geographical mobility and skills development)	Insight to identify transition pathways, see Box 5 below.
Jobseekers, incl. vulnerable groups	Employment incentives, skills-assessment, upskilling and reskilling, matching Job creation programmes	Recommender systems to nudge job search into sustainable occupations, online job matching Developing upskilling-reskilling programmes, including leading to micro-credentials
NEETs	VET with a focus on new occupations, informing on future skills needs	Aligning these training initiatives with the skills demanded in the labour market
Social entrepreneurs	Promoting self-employment and business start-up in green jobs	Financial incentives
Employees in all occupations	Career counselling with a focus on generic green skills useful in all occupations, on-the job learning, often with a focus on soft skills on environmental awareness	Pathways for Careers Training vouchers

Workers at risk of becoming unemployed

PES and partners can proactively identify workers affected by the green transition and customise services to meet their needs, reducing the risk of (long-term) unemployment, via tools identifying career transition pathways and studies.

Box 5. Tools to help with career transition[76]

France's Ministry for Ecological Transition and Solidarity and Ministry of Labour have developed **Supporting Career Transitions** website[77], a tool to identify occupations at risk of being displaced, as well as green and greening occupations and pathways for occupational transitions. It includes a comparison of tasks, competences, skills, qualifications, and wages.

Passport4Work[78], developed by a multi-level partnership collaboration of private companies, municipal services in the Eindhoven region in the Netherlands, the PES, and various social enterprises, is an online job-matching platform that aims to alleviate skills mismatches by increasing labour market transparency and providing highly personalised and gamified online counselling. The tool engages jobseekers in a gamified assessment to build their 'passport for work' (i.e. an overview of their skills and attributes).

PES have longstanding experience in mitigating the impacts of industrial restructuring. The energy transition and downsizing of fossil-based energy (such as coal mining) leads to job losses in these industries and indirectly affects other jobs through its impact on local economies. For this reason, occupational and regional mobility of workers and jobseekers need to be promoted, along with the development of local economies and social infrastructure. This calls for broader approaches, including using various labour market programmes and partnerships at sectoral and local levels.

Regions heavily affected by the energy transition have to cope with various challenges, such as the revitalisation of areas with persistent levels of high unemployment and poverty and social exclusion. These are best addressed in partnership with various actors[79]. PES activities include labour market analysis of affected local areas, support to local and regional stakeholders to diversify the regional economy, implementing upskilling and reskilling activities, and supporting job transitions. Evaluations have also shown the benefits of early intervention during the notice period by PES, before (individual or collective) dismissal takes place[80].

76 OECD, Job Creation and Local Economic Development 2023: Bridging the Great Green Divide, 2023, https://www.oecd.org/en/publications/job-creation-and-local-economic-development-2023_21db61c1-en.html.
77 Ministry for Ecological Transition (Ministère de la Transition écologique) https://www.ecologie.gouv.fr/politiques-publiques/lac-compagnement-transitions-profession-nelles#:~:text=Il%20s%E2%80%99est%20concr%C3%A9tis%C3%A9%20en%202015%20sous%20la%20forme%20d%E2%80%99un%20kit.
78 The Innovation in Politics Institute, Passport4Work, https://innovationinpolitics.eu/showroom/project/passport4work/.
79 European Network of Public Employment Services, PES support to greening of the labour market, Thematic paper, 2023, https://op.europa.eu/en/publication-detail/-/publication/04ec652e-1ae8-11ee-806b-01aa75ed71a1.
80 OECD, OECD Employment Outlook 2018, https://www.oecd.org/en/publications/oecd-employment-outlook-2018_empl_outlook-2018-en.html.

Examples: PES activities in regions affected by coal mining closures

To support structural change in the regions affected by the coal phase-out since 2020, the **German** federal government has earmarked significant funding to support the transition[81]. The research institute of the Federal Employment Agency (IAB) undertook a labour market analysis of the Rhenian region and established a qualification and occupational profile of mine workers (mainly VET-trained). One persistent challenge is to understand the full impact of the coal exit for the regional economy. The three local PES in the Rhenian Revier are working together[82], with the Revier PES agency tasked with coordinating activities with other actors. The companies and employees in the region need advice on change management and transition, individual funding opportunities, and offers to adapt qualifications to new economic structures. Concrete PES activities in the wider region included moderating dialogue between key companies in the summer of 2022 to foster cooperation on the design of transitions, a job hub, and creating perspectives[83].

In **Poland**, EU funding will support a broad set of measures to strengthen and diversify the regional economy, as well as improving quality of life in the region of Silesia. The Just Transition Fund is primarily intended to generate new jobs for people leaving the mining industry and related industries[84]. It supports the retraining of employees of large mining companies if the company's activities diversify and move away from mining activities towards a green and digital economy.

Spain's trade unions and the government agreed EUR 250 million for the *Plan del Carbón* to mitigate the effects of the closure of all Spanish coal mines. Some budget has been set aside to restore and environmentally regenerate former mining sites, with former miners given priority for employment in these jobs. An action plan will be created for each mining community, including plans for developing renewable energy, improving energy efficiency, and invest-ing/developing new industries[85].

In **Greece's** Western Macedonia and Peloponnese regions, coal was massively used in production, resulting in a special programme in March 2025 aimed at hiring 1 000 unemployed people affected by the transition to the post-coal era. A special pre-work programme was set up for 3 000 unemployed people aged 18-29 to enable them to gain work experience in the public and private sector. A special team of 10 PES staff was formed to implement these specific programmes.

Source: Taxonomy Working Group responses to PES Network survey.

81 German Federal Government, The Future Has Just Begun – A Short Profile of the Rhenish Lignite Area, 31 March 2021.
82 Rhenish Revier, https://www.arbeitsagentur.de/vor-ort/bruehl/rheinisches-revier.
83 European Network of Public Employment Services, PES support to greening of the labour market, Thematic paper, 2023, https://op.europa.eu/en/publication-detail/-/publication/04ec652e-1ae8-11ee-806b-01aa75ed71a1.
84 Through the Just Transition Mechanism, support will be available to all Member States, focused on carbon-intensive regions, or those with the most people working in fossil fuels.
85 Industriall Union, Spanish coal unions win landmark Just Transition deal, Press release, 1 November 2018, https://www.industriall-union.org/spanish-coal-unions-win-landmark-just-transition-deal/.

Measures implemented by PES in the context of employment losses can focus on promoting the transition of workers to other sectors and occupations. In the case of the restructuring of the **oil shale sector in Estonia**, the PES is designing several measures to support Just T ransition: salary grants for employers to shift from a carbon-intensive sector to another industry; degree study allowances to enable employees who were working in the carbon-intensive sector to acquire a new profession at a higher or vocational education institution; micro-qualifications (micro-credentials) allowance to enable employees who were working in the oil shale industry to gain new qualifications[86]. Another example from Sweden showcases support for mobility of the labour force to regions where jobs are created in the green economy.

Example: PES activities to support regional and occupational mobility towards dynamic greening industries in Sweden

Sweden is seeing new and large investments and expansions in renewable energy, especially in the northern parts of the country (worth an estimated EUR 100 billion). Upper Norrland is a key mining region, with the extraction of rare minerals needed for the clean transition. A regional PES office supports the new industrialisation strategy via detecting skills needs through surveys, spreading information about skills needs in the expanding industries (mainly in Västerbotten and Norrbotten), stimulating geographical and professional mobility, and strengthening associated strategic collaboration[87]. The PES has developed an analysis model to define large business investments and expansions.

Source: PES Network, 2023; OECD 2021; information provided by the Swedish PES to the PES Network.

Support for jobseekers, including vulnerable groups and NEETS

PES have primarily been involved in implementing small-scale employment incentives in the context of the green transition[88]. In some cases, they have also been involved in their design, alongside line ministries and other ministries. Measures focus mainly on green sectors. Employment subsidies, for example, aim to support hiring unemployed people in green jobs (e.g. the government in **Slovenia** launched the 'Green Jobs' employment incentive in 2021[89], in cooperation with the Ministry of the Environment, Climate and Energy, while **Bulgaria** has had a small-scale dedicated employment incentive programme for more than a decade).

In **Greece**, a programme for 5 000 unemployed people aged 18-66, focused on the green economy sector and funded by the Recovery and Resilience Facility (RRF) was set up to support full-time employment in businesses that are active, or will be active, in the green economy.

86 European Network of Public Employment Services, PES support to greening of the labour market, Thematic paper, 2023, https://op.europa.eu/en/publication-detail/-/publication/04ec652e-1ae8-11ee-806b-01aa75ed71a1.
87 Swedish PES, Uppdrag om kompetensförsörjning vid stora företagsetableringar och företagsexpansioner, 18 December 2021.
88 See for more examples: European Network of Public Employment Services, PES support to greening of the labour market, Thematic paper, 2023, https://op.europa.eu/en/publication-detail/-/publication/04ec652e-1ae8-11ee-806b-01aa75ed71a1.
89 'Green Jobs' employment incentive is an ALMP programme where only employers who have hired an unemployed person within a specific occupation in a particular sector, selected by the Ministry of the Environment, Climate and Energy, were eligible to receive a subsidy. The selection of occupations and sectors was not based on any common definition.

An example funded by EEA and Norway Grants Fund which trains NEETS in green skills is described below:

Example: Training NEETS through the INTERCEPT project

The INTERCEPT project aimed to harness the potential of NEETs aged 25-29 in the green economy. It brought together universities, research and training institutions, analysis and planning institutes, as well as employment promotion agencies, from six European countries: **Italy, Lithuania, Luxembourg, Malta, Poland and Slovakia**. The project conducted comprehensive research on the demographics, trends and needs of NEETS, as well as the skills employers deem necessary for green jobs. It also offered a 25-hour training opportunity in green skills and facilitated 240-hour work placements with green companies in Italy, Malta and Lithuania. INTERCEPT led to the development of an introductory green skills training programme – Award in Green Jobs Motivational Training – marking an important step towards addressing the evolving needs of Malta's green transition. This training programme is accredited by the Malta Further and Higher Education Authority (MFHEA) and is available for free to working age individuals residing in Malta through Jobsplus. By aligning these training initiatives with the skills demanded by emerging and evolving industries, INTERCEPT established a foundation for creating new job opportunities within the green economy.

More information is available at **https://interceptproject.eu/**

Job creation programmes **target disadvantaged groups** (e.g. NEETs, long-term unemployed, recipients of social benefits and other vulnerable groups) with a focus on projects that are socially and environmentally useful, without competing with the regular labour market. However, examples of the dedicated use of employment incentives targeting both vulnerable groups and the green economy are limited for several reasons:

- Many jobs in the green economy require skills at a medium and high qualification level that people in vulnerable groups often do not possess.
- Employment incentives used in the green economy, for example facilitating the labour market transition of specific groups such as young people or older people, may happen more often but not be recorded by the PES as a specific employment incentive(s) for green jobs.

Social entrepreneurs

Social enterprises and cooperatives also support projects on green transition. In **France**, regional partnerships support the PES' activities to place long-term unemployed people into sustainability projects implemented by social enterprises (e.g. Territoires zéro chômeur de longue durée, implemented in some regions[90]). More generally, social economy organisations in France are often at the forefront of the green transition because their social objectives align with environmental sustainability.

90 Territoires zéro chômeur de longue durée, https://www.youtube.com/watch?v=xLspaQU2vIM&feature=emb_title.

In **Croatia**, in 2025, unemployed individuals can receive support of up to EUR 20 000 to start their own business in the digital and green sectors. This includes funding to cover the costs of establishing a business, with a requirement to work in the business full-time for at least 24 months and provide evidence of eligible costs. Green jobs are defined as those providing environmental benefits, assessing compliance with environmental standards, educating on green technologies, and aiding green business transformation.

The Empreende XXI in **Portugal** provides start-up funding for investment through a non-repayable grant up to a limit of 40% of the eligible investment, increased by 15% for new products or service in the areas of technology, climate transition, business model, among others.

Employees in all occupations

PES initiatives focus also on generic green skills and skill enhancements for all occupations to transition to greener practices. For instance, PES are partners in programmes to support the green transition within a sector, such as in **Sweden**. The PES supported a regional project, co-financed by the ESF, providing the competitive food production industry with a skilled workforce through skills development initiatives and targeted job-matching[91]. In **Slovenia**, the Catalogue of Training and Education Programmes includes 192 programmes. These were reviewed in 2023 to target digital and green skills, with the assistance of registered external training providers. Where possible, external providers were invited to supplement their programmes with digital or green components. As a result, certain programmes were updated to include this content, if it was not already present. Following the revision, programmes containing digital and/or green elements were defined and appropriately marked. Out of 192 programmes listed in the Catalogue of Training and Education Programmes, as many as 48 programmes – or 25% – are classified as programmes with green elements. Programmes with green elements refer to training or education programmes that include content related to environmental sustainability, climate action, or the green transition.

DESIGNING SERVICES AND MEASURES: LESSONS AND CONSIDERATIONS FOR PES

Key takeaways



PES have already several services in place to support clients during the green transition. Often services focus on promoting employment in green jobs and equipping jobseekers and workers at risk of losing their jobs with the necessary skills.

A 'skills-based' approach across different PES customer groups, from unemployed to employees across all occupations, ensures that people are equipped with relevant and changing skills needs the green transition requires. This is especially relevant for transferable skills to support job-to-job transitions. For the unemployed, some PES design small-scale employment incentives for the green transition, and social entrepreneurs offer employment opportunities. However, skill requirements of green jobs are a challenge for job creation for disadvantaged groups.

91 European Network of Public Employment Services, PES support to greening of the labour market, Thematic paper, 2023, https://op.europa.eu/en/publication-detail/-/publication/04ec652e-1ae8-11ee-806b-01aa75ed71a1.

Lessons learned, considerations and recommendations	Examples
Identifying workers at risk and supporting regional transitions. Supporting for job-to-job transitions for people at risk of losing their job, such as via working in partnership with employers who face skills shortages, offer holistic services including reskilling or upskilling, as well as relocation services.	See examples from Germany, Greece, Poland, Spain in Section 3.2 Swedish Relocate programme in Section 3.2. For more advanced PES with capacity for data analytics: consider mapping career transitions.
Supporting jobseekers, including vulnerable groups, via upskilling in generic and specialised green skills.	Examples in Section 3.2.
Promoting green entrepreneurship, as their social aims often align sustainability targets. PES can support business plan development with a green focus.	Examples in Section 3.2

4 WORKING WITH EMPLOYERS

Employers need a labour force that is equipped with skills for the green transition and green skills shortages have been reported in a large majority of countries[92]. Skills shortages span a range of industries and sectors, with the most frequently cited being renewable energy (photovoltaic), construction, engineering, electrical engineering and mechatronics (in descending order), alongside other industries. Consistent collaboration with employers across PES responsibilities ensures that services to employers are relevant to meet workforce demand. For instance, the taxonomy by PES France is available to employers who are mandated to list the required skills for vacancies they post online, while employers are mandated to list the required skills for vacancies they post online. When working with different employers, PES may want to consider different segments and types. The 'Eco-Transformation of Industries Matrix' created by the ILO (and advanced by the UK charity NESTA[93]) categorises distinguishes employers based on two criteria: carbon emission levels and their environmental initiatives.

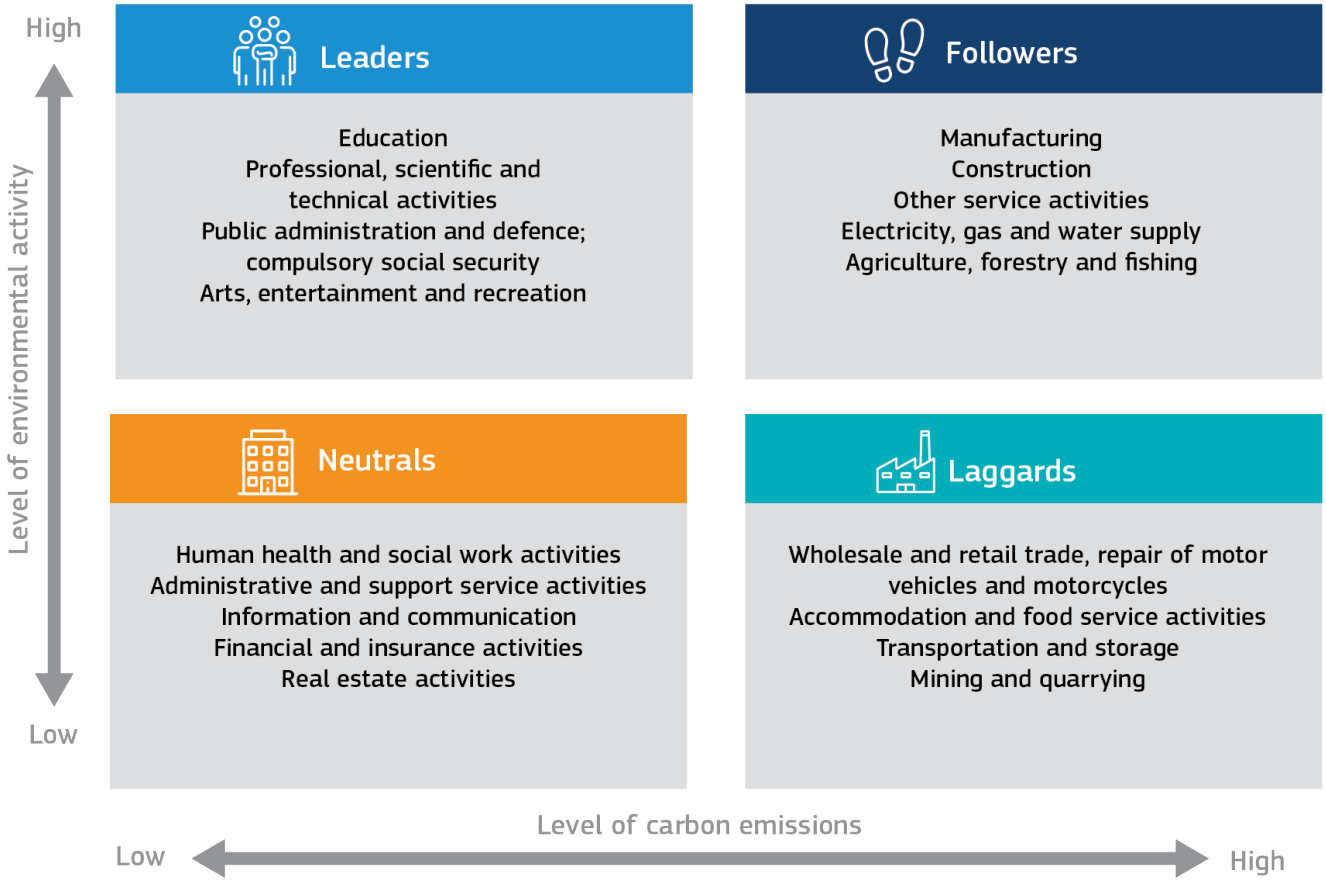
- **Leaders:** Industries that are most eco-friendly, as they do not produce high levels of carbon emissions and are intensively involved in activities that directly protect the environment across the economy;
- **Neutrals:** Industries in this category produce low levels of carbon emissions but are not involved in activities that directly protect the environment. They are part of the green sector but are not influenced by new climate-crisis policies;

92 McGrath, C., Heather Rose, Souto-Otero, M., Report on Green Skills: Network of Experts in Adult Learning, Summary Report, 2022.

93 Nesta, Going Green, Preparing the UK workforce for the transition to a net-zero economy, 2020, https://media.nesta.org.uk/documents/Going_Green-_Preparing_the_UK_workforce_to_the_transition_to_a_net_zero_economy.June.2020.pdf.

- **Followers:** Although producing high levels of emissions, followers are also intensively involved in activities that are intended to protect the environment and could thus create green jobs;
- **Laggards:** Industries in this category produce high levels of carbon emissions and are not involved in activities to protect the environment.

Figure 5. The 'Eco-Transformation of Industries Matrix'



Source: Nesta, Going Green, Preparing the UK workforce for the transition to a net-zero economy, 2020,

For PES, collaboration with leaders entails that the most sustainable industries need to be identified via labour market analysis, supporting employers via the development of skills profiles and job descriptions, and tailoring their courses to reflect this need.

Box 6. Working with employers to identify green skills and tailor training

Method	Examples
Surveys and studies	PES France published a survey of 15 000 employers in 2022, showing that 4 in 10 employers needed new working methods or new skills for the green transition[94].
Development of skills profiles	PES Sweden works closely with employers to harmonise their skills profile requests for workers. For example, the Northvolt battery factory in Skellefteå has a high demand for battery production workers. Both public and private vocational training providers support workers with the necessary skills in battery production, but also find it challenging to establish the skill profile for battery production workers (e.g. **maintenance technician**) because the profile has changed as production has developed.
Sectoral approaches	In Germany, various regional labour market actors collaborate toenhance VET, ensuring a supply of skilled workers. These networks include cooperation on identifying skill needs and forming continuous training partnerships among companies, training providers and the PES to address specific sector requirements. Continuous training agencies (Weiterbildungsagentur) serve as one-stop-shops providing transparent access to training and qualification offers for citizens and employers. Key participants are employers, regions, municipalities, educational providers, chambers, associations, trade unions, and employment agencies.

94 PES France, Recrutement, compétences et transition écologique ; des enjeux qui se polarisent sur quelques secteurs - France Travail, https://www.francetravail.org/statistiques-analyses/entreprises/offres-demploi-et-recrutements/recrutement-competences-et-transition-ecologique--des-enjeux-qui-se-polarisent-sur-quelques-secteurs.html?type=article.

Analysis for training needs	In Portugal, within the Pro_Mov project, companies are actively involved in the entire training process, from the analysis of training needs and design of tailored programmes to the selection of candidates and the integration of work-based training and hiring. The PES supports candidates to transition to a new profession after receiving tailored training and mentoring. The project already has a wide range of training activities in seven different laboratories, including the Green Jobs laboratory (training professionals in microgeneration and electric mobility). Since 2024 the Austrian PES offers employers an Organisational Impulse Consulting (IBB-Impulsberatung für Betriebe) to the subject of ecological tructural change. The programme provides HR consulting services and is carried out by external contractors through a standardised consulting process (four phases), which help to translate pressing HR issues into sustainable labour market solutions. Measures are designed and implemented to adapt or improve a company's HR Man-agement concerning the ecological transformation, also with funding or services offered by the Austrian PES, e.g. to train or upgrade employees. Since 2022, the Austrian PES runs the Environmental Foundation programme that offers workplace-based training (tailored training and internships in companies) for unemployed in the environmental sector. Participants acquire full apprenticeship diplomas in a shortened training period, or those with existing apprenticeship diplomas or university entrance qualification acquire higher qualifications. The advantages for employers are that they can fill vacancies, the qualification is promoted by the PES and they have no wage labour costs during the training period (as this training is an internship).

Source: European PES Capacity Report, 2023; questionnaire responses to PES Network taskforce; PES Network, 2023.

"Followers" and "laggards" are often active in brown sectors, as they are associated with high carbon emissions. Examples of PES working together with "followers" and "laggards" are de-scribed in Section 3.2. Here, participation in on-the-job learning is key. One example outside of the EU is the approach taken in the UK where the workforce in routine and manual jobs is mostly composed of workers with lower qualifications, requiring new skills. Technical and soft skills described required in the green transition need to be enhanced, especially as participation in lifelong learning is much lower in "laggard" industries (in the UK, participation rate of leader employees in adult learning is about 21%, compared to only 11% for "laggard" employees[95]).

95 Nesta, Going Green, Preparing the UK workforce for the transition to a net-zero economy, 2020, https://media.nesta.org.uk/documents/Going_Green-_Preparing_the_UK_workforce_to_the_transition_to_a_net_zero_economy.June.2020.pdf.

WORKING WITH EMPLOYERS: LESSONS AND CONSIDERATIONS FOR PES

Key takeaways



Categorising employers with regards to their relevance to the green economy, can help to target PES employer services better. Ideally, PES should work with employers to implement a 'skills-based' approach by assessing skills needs and facilitating relevant training needs in the context of the green transition. To ensure relevant training, employers can be involved via setting up workplace-based learning or inputting into classroom training.

Lessons learned, considerations and recommendations	Examples
Segmenting employers into different categories to customise the collaboration.	See Eco-Transformation of Industries Matrix[96]
Collaborating to identify skills needs via surveys and studies, local or sectoral forums.	See Box 6. Working with employers
Building on social corporate responsibility willing to employ vulnerable jobseekers.	The French PES involves companies in recruiting people distant from the labour market, notably in green jobs. The club 'Companies are committed' (Les entreprises s'engagent[97]) brings together companies in 200 local clubs.
Supporting employers to develop skills profiles for shortage professions.	PES Sweden works with employers to develop skills profiles, in Section 3.2 . PES Slovakia consults with employers in individual sectors
Cooperating with employers to offer workplace training.	PES Austria in Box 6. Working with employers

96 Ibid.
97 https://lesentreprises-sengagent.gouv.fr/les-entreprises-engagees

5. CONCLUSIONS

1 This Toolkit shows the importance of PES working in partnership to identify skills needed for the green transition, develop skills measures, and support regional and occupational mobility of jobseekers. This includes also raising awareness and encouraging women to enrol and work in technical (STEM) fields and IT, which are in demand for 'greening' labour markets. At the local level, PES can become active partners in managing small-scale local ecological transition projects. Partnerships would also include setting up cooperations with local key stakeholders, such as employers, social partners or training providers, to manage the large-scale employment impact of industrial restructuring linked to the ecological transition. Cooperation with key stakeholders is also essential to integrate disadvantaged groups into the labour market, including in activities related to the green economy. Key partners include employers, sectoral organisations, training institutions, and municipalities.

2 The Toolkit includes guidance on identifying the impact of the green transition at regional and local level, as well as at sectoral level. The labour market impact varies considerably between local areas, given the differences in local economic structures.

3 Green jobs and skills will continue to change and develop. Accordingly, PES need to constantly inform themselves or conduct labour market analyses to grasp the changes in occupational and skills profiles.

4 The PES Network should continue to collect specific sectoral and occupational practices, as well as, for instance, good practices on attracting women into green sectors (where evidence remains scarce). It will continue its work on a common definition of green jobs, while acknowledging persist differences between countries.

ANNEX 1 OVERVIEW OF PES' APPROACHES IN DEFINING GREEN JOBS.

Definitions of green jobs and related concepts used by the French PES using an occupation-based approach

In **France**, the **National Observatory for Jobs and Occupations of the Green Economy** (Onemev) was created in 2010, bringing together a wide range of actors, including the Ministry of Employment and the French PES[98]. Using a task-based, job-based and occupation-based approach, Onemev identified **10 green occupations and about 52 greening occupations based on** the PES classification of ROME occupations[99]. Onemev departs from the definition established by Eurostat and the OECD.

It looks more closely at certain activities: research and development (R&D), environmental engineering, and certain public services. In addition, the approach includes 'peripheral activities' that contribute to better environmental quality but less directly to the core of the green economy (Cedefop, 2019).

The following definition is shared by all national institutions (and is not specific to the PES):

'**A green job** is one whose purpose and most of its associated skills contribute to measuring, preventing, controlling and correcting negative impacts and damage to the environment. More specifically, it aims to achieve at least one of the following objectives:

- reduce energy and raw materials consumption,
- limit and reduce GHG emissions,
- minimise waste and pollution,
- protecting and restoring ecosystems,
- helping businesses and communities adapt to climate change.'

A **'greening job'** is defined as 'one whose purpose is not environmental, but whose associated skills are evolving in line with the need to develop know-how and to integrate new activities and/or technologies that reduce the environmental impact of the activity'. Data are gathered through a centralised statistical entity. In addition to green and greening jobs, France Travail also identifies **'jobs linked to the ecological transition'**: these are not necessarily green or greening jobs, but jobs that are crucial to the ecological transition trajectory (e.g. bicycle repairer - the associated skills are not green per se, but this job is essential for implementing eco-friendly mobility solutions).

The definition used by the French PES is not based on international standards. Nevertheless, O*NET served as inspiration during the development of the occupational classification used by the French PES.

The implementation of ROME 4.0 in early 2023 allowed for an enriched approach to green jobs, which should be in full effect by the end of 2024. Other organisations involved in identifying labour market evolutions in light of the green transition include the Directorate for Research, Studies and Statistics (DARES) and the National Institute of Statistics and Economic Studies (INSEE).

98 Onemev, https://www.ecologie.gouv.fr/lobservatoire-national-des-emplois-et-metiers-leconomie-verte.
99 As the PES classification does not allow a quantification of jobs, these occupations were translated into the usual classification. This means that nine green occupations and around 70 greening occupations have been identified in the national classification system of occupations, https://www.notre-environnement.gouv.fr/site-ree/themes-ree/economie-verte/emplois/les-emplois/article/les-emplois-dans-les-metiers-de-l-economie-verte.

Source: Taxonomy Working Group responses to PES Network survey; questionnaire responses to the PES Network; PES Network, 2023.

Definition of green occupations by the Polish PES

In Poland, green occupations are defined for PES purposes in the absence of a nationwide definition.

Green occupations are defined as specialties related to broadly understood activities in the fields of environmental protection, ecology, renewable energy sources and nuclear energy, public transport sector, ecological construction, waste management, as well as services important for the quality of life (including care as it is broadly understood).
A list of green occupations has been developed in the Polish Classification of Occupations and Specialties, but is a proxy for skills and is not binding.

Work is underway to implement the Classification of Skills for PES purposes, notably as part of the project 'Development and implementation of a strategic forecasting system for skills demand in the area of green and digital economy'.

Source: Taxonomy Working Group responses to the PES Network survey.

Defining and monitoring green jobs and green transition by the German PES

The Federal Employment Agency (BA) takes two different approaches to monitoring the green transition:

1) The IAB has developed the so-called **Greenness-of-Jobs Index (GOJI)**, an occupational index that describes the shares of green skills and brown skills[100] relative to the total number of skills per occupation. It is based on the BA's online database (BERUFENET). The GOJI presents percentages of potentially environmentally friendly tasks: Renewable energy technology, Business organisation and strategy; Plumbing and heating, Technical railroad operation, Building services engineering. (e.g. it enables the observation of change over time in official occupational descriptions from the online database). It is available at a five-digit level of the German Occupational Classification (KldB2010) and is transferable to the International Standard Classifica-tion of Occupations (ISCO). The GOJI enables aggregation of data at multiple levels: occupational, geographical, sector, company/establishment, and worker. It can be broken down at local level to show the spatial distribution of greenness of jobs in Germany. The GOJI is updated yearly, from 2012 until the most recently available data.

2) Statistical information provided by the Statistics Department of the Federal Employment Agency: **The Statistics Department monitors three main sectors - environmental protection, energy and transportation** because major transformation projects (e.g. investing in renewable energy sources, phasing out coal, renovating buildings to make them more energy-efficient and changing heating systems, electromobility and the transport transition) generally target these areas.

100 Examples for Brown Skills: plastics processing, concrete technology, iron weaving, dairy farming and production, Firing (ceramics, enamel), glass melting, leather processing, steel construction, flight tourism.

Source: Taxonomy Working Group responses to the PES Network survey; Janser, M. (2019). The greening of jobs: Empirical studies on the relationship between environmental sustainability and the labor market (Doctoral dissertation, Otto-Friedrich-Universität Bamberg, Fakultät Sozial-und Wirtschaftswissenschaften). and Bachmann, R., Janser, M., Lehmer, F., & Vonnahme, C. (2024). Disentangling the greening of the la-bour market: The role of changing occupations and worker flows. IAB Discussion Paper 12/2024.

Definition of green jobs triggered by the implementation of new ALMPs in Croatia

Since 2023, new ALMPs are specifically designed to support employment in green (and digital) jobs. The PES issued a manual to assess whether a job satisfies the criteria to be considered green. Similarly, within the voucher training system, the Ministry of Labour, Pension System, Family and Social Policy introduced a skills catalogue to determine whether a training programme is connected to green skills. The definition has been developed by the Croatian PES according to Directive (EU) 2020/852, together with definitions of green skills in the ESCO and the HKO. Croatia considers that jobs containing green skills are defined as green occupations.

Source: Taxonomy Working Group responses to the PES Network survey; PES Network, 2023.

ANNEX 2 ACTIVITIES BY PES TO ADDRESS THE IMPACTS OF THE GREEN TRANSITION

The table below provides a comparative overview of the activities undertaken by PES across various European countries to address the impacts of the green transition on the labour market. Many PES across Europe (e.g. Belgium (Actiris, Forem, VDAB), Denmark, Germany, Austria) have already implemented comprehensive measures, such as analyses of labour market trends, initiatives to mitigate negative effects for jobseekers, or upskilling and reskilling supports. Other PES, including Bulgaria, France and Poland, are in the planning stages of such activities, suggesting that mutual learning exercises, a strengthened network and toolkits could help PES to learn from one another. Several countries (Greece, Latvia, Iceland) show limited engagement in mitigating job losses but still provide support through career guidance and ALMPs.

By contrast, countries such as Lithuania display fewer initiatives overall, indicating potential to strengthen their support systems for jobseekers. Very few countries have specifically developed programmes for informing young people about future skills needs (France, Malta) or promoting self-employment and green start-ups (Lithuania). Specifically, business creation was supported if the business idea aligned with sustainability goals.

Early intervention to support those at risk of losing their employment and integrating disadvantaged groups into the labour market through green initiatives are areas that could be explored by PES.

Table A.1: Overview of the activities undertaken by PES across various European countries to address the impacts of the green transition on the labour market

PES	Analyses of labour market trends	Mitigate negative effects linked to job losses	Supporting jobseekers to adapt their skills, upskill, retrain and engage in green VET
Belgium - Actiris	Dependent on future government decision	No	Training offer
Belgium - Forem	Yes	Yes	Career guidance; ALMP; information
Belgium - VDAB	Yes	No	Career guidance; ALMP; information
Bulgaria	Plans to do so in the future	Yes	Career guidance; ALMP; information
Denmark	Yes	No	Career guidance; ALMP; information
Germany	Yes	Yes	Career guidance; ALMP; information
Estonia	Plans to do so in the future	Yes	Career guidance; ALMP; information
Greece	Yes	No	Career guidance; ALMP; information
Spain	Plans to do so in the future	No	Career guidance; ALMP; information
France	Plans to do so in the future	No	Career guidance; ALMP; information
Croatia	Yes	No	Career guidance; ALMP; information
Latvia	No	No	No
Lithuania	No	No	Career guidance
Luxembourg	No	No	ALMP
Austria	Yes, improvements still needed and ongoing	Yes	Career guidance; ALMP; information
Poland	Plans to do so in the future	Yes	Career guidance; ALMP; information
Slovenia	Plans to do so in the future	Yes	Career guidance; ALMP; information
Slovakia	No	No	ALMP
Sweden	Plans to do so in the future	No	ALMP
Norway	Yes		Career guidance; ALMP; information
Iceland	No		Career guidance; ALMP; information

ANNEX 3 OVERVIEW OF RESOURCES FOR PES

Studies	
Cedefop	Skills forecast
Cedefop	Skills intelligence
Cedefop	Green observatory
Cedefop	Tracking the green transition in labour markets
Cedefop	The importance of 'thyroid occupations'
PES Network	Greening of the labour market Impacts for PES
Nesta	Going Green
Datasets	
Cedefop	Skills-OVATE
European Commission	European Classification of Occupations, Skills, and Competences (ESCO)
Eurostat	Environmental goods and services sector (EGSS)
International Labour Organisation	International Standard Classification of Occupations (ISCO)
Occupational Information Network	O*NET® 29.1 Database
PES Strategies and partnerships	
Austria	Environmental Foundation Programme's workplace-based training
Belgium-Forem	Internal sustainability strategies focusing on energy consumption, sustainable transport, and public procurement policy
Belgium-Forem	Project to identify the occupations of the future for Wallonia
Croatia	Identifying skills for the green transition via regular meetings
France	One-hour online training on green and greening occupations for PES staff
France	Identification of competences required for the circular economy
France	Green section in MétierScope
France	Tool to identify occupations at risk of being displaced and green occupations
France	Regional partnerships like Territoires zéro chômeur de longue durée

France	Les entreprises s'engagent for recruiting people distant from the labour market in green jobs
Germany	Green section in BERUFENET
Germany	The future has just begun for regions affected by coal mining closures
Netherlands	Passport4Work online job-matching platform
Norway	Projects with universities to improve PES employees' knowledge of markets
Poland	ESF+ funded project to set up a forecasting system including green skills
Poland	Development and implementation of a strategic forecasting system for skills demand in the green and digital economy
Poland	Measures to strengthen and diversify the regional economy in Silesia
Portugal	Refurbishment of VET Centres with technologically intensive equipment for green professions
Portugal	Support for entrepreneurship for innovative solutions
Slovenia	Strategy on Greening of the Labour Market in its Sustainability Report
Spain	Plan del Carbón to mitigate the effects of coal mine closures
Sweden	Support for regional and occupational mobility towards greening industries
Sweden	Green jobs initiative
Malta, Italy, Lithuania	INTERCEPT project

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